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# CARBOXYLIC ACIDS & DERIVATIVES

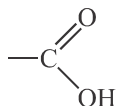
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| Reactivity    | Derivative                                                                                                                | Leaving Group                                                         | Basicity   |
|---------------|---------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------|------------|
| More Reactive | Acid Chloride<br>$\text{R}-\overset{\text{O}}{\parallel}{\text{C}}-\text{Cl}$                                             | $\text{Cl}^{\ominus}$                                                 | Less Basic |
|               | Anhydride<br>$\text{R}-\overset{\text{O}}{\parallel}{\text{C}}-\text{O}-\overset{\text{O}}{\parallel}{\text{C}}-\text{R}$ | $^{\ominus}\text{O}-\overset{\text{O}}{\parallel}{\text{C}}-\text{R}$ |            |
|               | Ester<br>$\text{R}-\overset{\text{O}}{\parallel}{\text{C}}-\text{O}-\text{R}'$                                            | $^{\ominus}\text{O}-\text{R}'$                                        |            |
| Less Reactive | Carboxylate<br>$\text{R}-\overset{\text{O}}{\parallel}{\text{C}}-\text{NH}_2$                                             | $^{\ominus}\text{NH}_2$                                               | More Basic |

## THEORY

## 1. INTRODUCTION

**Carboxylic acids** are the compounds containing the **carboxyl functional group**.



The carboxylic acids may be **aliphatic** ( $R-COOH$ ) or **aromatic** ( $Ar-COOH$ ) depending upon whether  $-COOH$  group is attached to aliphatic alkyl chain or aryl group respectively.

The general formula for carboxylic acids is  $C_nH_{2n}O_2$ .

Carboxylic acids are used in the production of polymers, pharmaceuticals, solvents, and food additives. Industrially important acids include acetic acid (component of vinegar, precursor to solvents and coatings), acrylic and methacrylic acids (pre cursors to polymers, adhesives), adipic acid (polymers), citric acid (beverages), ethylene diaminetetraacetic acid (chelating agent), fatty acids (coatings), maleic acid (polymers), propionic acid (food preservatives), terephthalic acid (polymers). Esters of fatty acids are the main components of lipids and polyamides of aminocarboxylic acids are the main components of proteins.

The carboxylic acids ionize in water to give  $H^+$  ions and hence they are acidic in nature.

## Important Aliphatic Acids

|               |                                     |
|---------------|-------------------------------------|
| $HCOOH$       | Formic Acid                         |
| $CH_3COOH$    | Acetic Acid                         |
| $CH_2=CHCOOH$ | Acrylic Acid                        |
| $CH_3COCOOH$  | Pyruvic Acid ( $\alpha$ -keto acid) |

## Important Aromatic Acids

|                |                  |
|----------------|------------------|
|                |                  |
| Benzoic Acid   | Phthalic Acid    |
|                |                  |
| Salicylic Acid | Anthranilic Acid |

These are further classified as mono, di, tri carboxylic acids depending upon number of carboxylic groups present in their molecules.

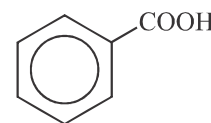
## Mono Carboxylic Acids



Formic Acid



Acetic Acid

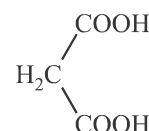


Benzoic Acid

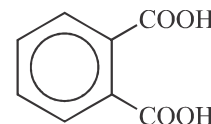
## Di Carboxylic Acids



Oxalic Acid



Malonic Acid



Phthalic Acid

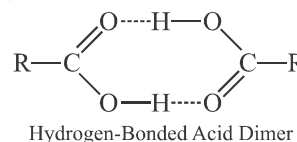
## 2. PHYSICAL PROPERTIES &amp; ACIDIC CHARACTER

## 2.1 Boiling Point

Carboxylic acids boil at considerably higher temperatures than do alcohols, ketones, or aldehydes of similar molecular weights.

|                               |                             |                                  |
|-------------------------------|-----------------------------|----------------------------------|
|                               |                             |                                  |
| Acetic Acid, bp $118^\circ C$ | 1-Propanol, bp $97^\circ C$ | Propionaldehyde, bp $49^\circ C$ |

The high boiling points of carboxylic acids result from formation of a stable, hydrogen-bonded dimer. This dimer contains an eight-membered ring joined by two hydrogen bonds, effectively doubling the molecular weight of the molecules leaving the liquid phase.



## 2.2 Melting Point and Physical State

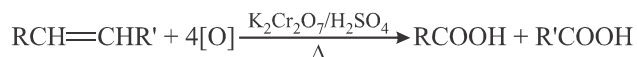
Aliphatic acids upto nine carbon atoms are colourless liquids at room temperature with unpleasant odours. The higher acids are wax-like solids and are practically odourless due to low volatility. They exist as dimer in aprotic and non-polar solvents.

Unbranched acids made up of an even number of carbon atoms have melting points higher than the odd numbered homologs having one more or one less carbon. This oscillating behaviour in melting point is explained by the differences in the intermolecular attractive forces in the crystalline state.

## 2.3 Solubility

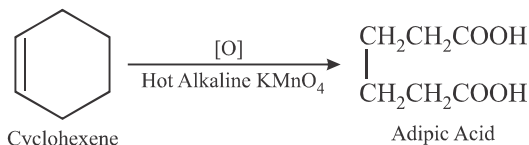
Carboxylic acids form hydrogen bonds with water, and the lower-molecular-weight carboxylic acids (up through 4 carbon atoms) are miscible with water. As the length of the hydrocarbon chain

Oxidising agents used are acidified or hot alkaline  $\text{KMnO}_4$  and  $\text{K}_2\text{Cr}_2\text{O}_7$ .



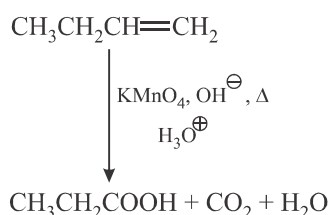
Cyclic alkenes can be oxidized to dicarboxylic acids.

### Example – 2

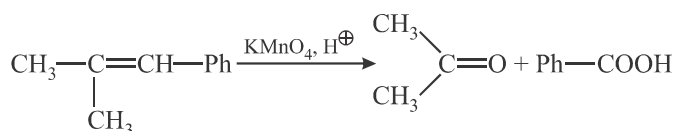


*Note...* 1. In oxidative cleavage, each C of multiple bond becomes the carbon atom of the carbonyl group and a hydrogen substituent on the double bond is replaced by a hydroxyl group.  
2. The terminal  $-CH_2$  group of 1-Alkene is completely oxidized to  $CO_2$  and  $H_2O$ .

### Example – 3



### Example – 4

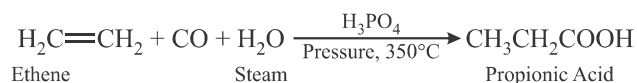


### 3.2.3 Kosch Reaction – Hydrocarbonylation

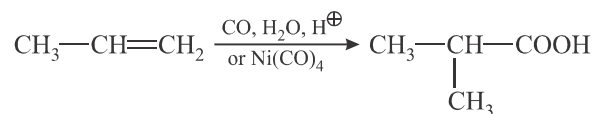
### Example – 5

Alkene on treatment with  $CO$ ,  $H_2O$  at high temperature and pressure causes carbonylation. Usually,  $H_3PO_4$  is used as the acid catalyst. The similar effect is observed when  $Ni(CO)_4$  is treated independently.

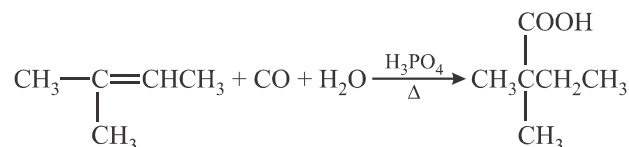
### Example – 6



### Example – 7



### Example – 8

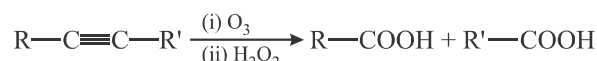
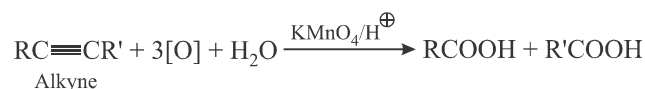


Number of carbons is one more than parent alkene.

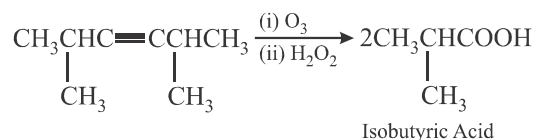
## 3.3 Alkynes

### 3.3.1 Ozonolysis

Alkynes on ozonolysis give carboxylic acids.



### Example – 9



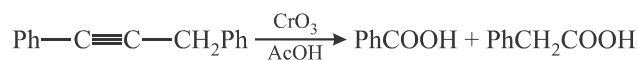
### 3.3.2 Oxidation

The use of strong oxidising agents such as  $KMnO_4/H^{\oplus}$ ,  $KMnO_4/OH^{\ominus}/\Delta$ ,  $CrO_3$  etc. directly gives the carboxylic acid.

### Example – 10



### Example – 11

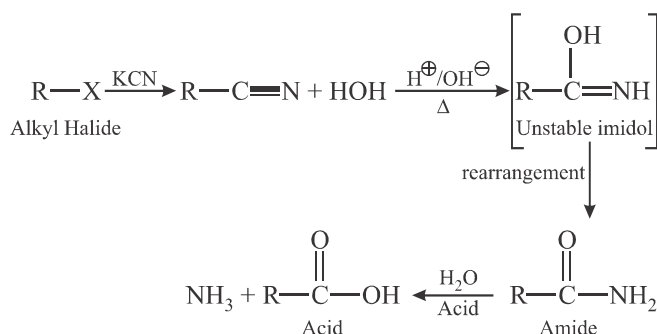


## Carboxylic Acids & Derivatives

5

### 3.4 Alkyl Halides

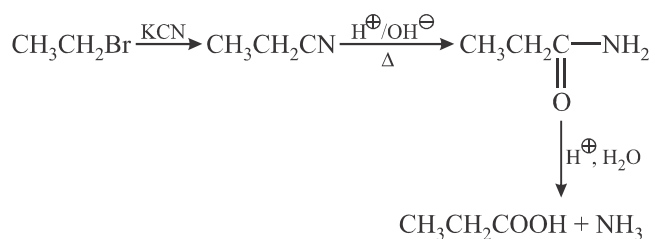
#### 3.4.1 KCN + Hydrolysis



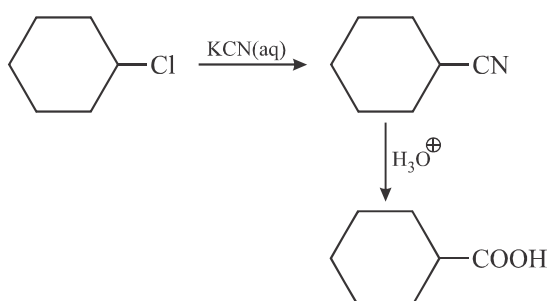
**Note...** 1. In acidic medium, the products are  $\text{RCOOH}$  and  $\text{NH}_4^+$  while in alkaline hydrolysis,  $\text{RCOO}^-$  and  $\text{NH}_3$  are products.

- The acid formed contains one more C than parent alkyl halide. The order of reactivity of alkyl halides is  $1^\circ > 2^\circ > 3^\circ$ .
- In case of  $3^\circ$  alkyl halide, elimination takes place.
- Vinylic, allylic halides are not appropriate as substitution is difficult.

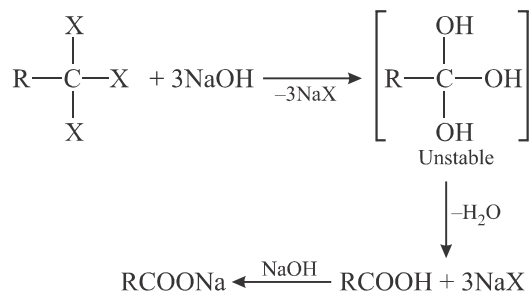
#### Example – 12



#### Example – 13

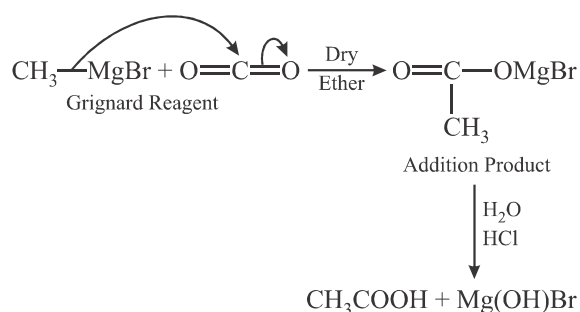


#### 3.4.2 Hydrolysis of terminal trihalogen derivatives of Alkanes



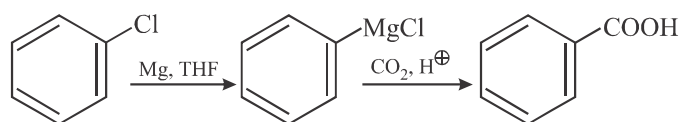
#### 3.4.3 Grignard Reagent + $\text{CO}_2$

##### Example – 14



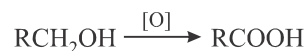
Carboxylic acid formed has one C atom more than Grignard Reagent.

##### Example – 15



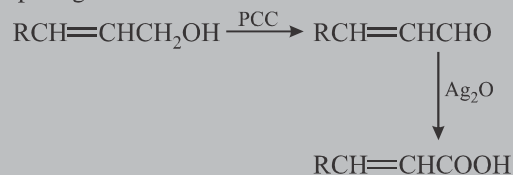
### 3.5 Alcohols

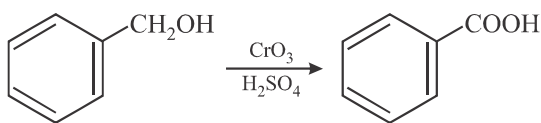
Oxidation of  $1^\circ$  alcohols yields carboxylic acids.



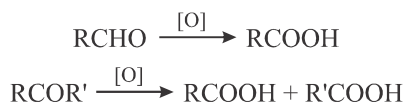
**Note...** 1. **Strong Oxidizing Agents :**  $\text{K}_2\text{Cr}_2\text{O}_7/\text{H}^+$  or  $\text{KMnO}_4/\text{H}^+$ .

2. **Mild Oxidising Agents :**  $\text{MnO}$ ,  $\text{CuO}$  (Fehling's Solution),  $(\text{CrO}_3, \text{H}_2\text{SO}_4)$  in acetone solution,  $\text{Ag}_2\text{O}$  (Tollen's reagent).  $\text{FeSO}_4 + \text{H}_2\text{O}_2$  (Fenton's reagent). These are used for preparing unsaturated acids.

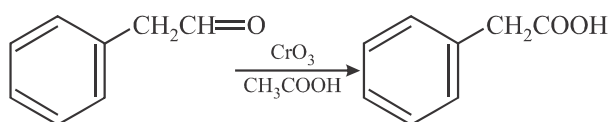


**Example – 16**

**3.6 Aldehydes and Ketones**
**3.6.1 Oxidation**

Oxidation of aldehydes and ketones give carboxylic acids.



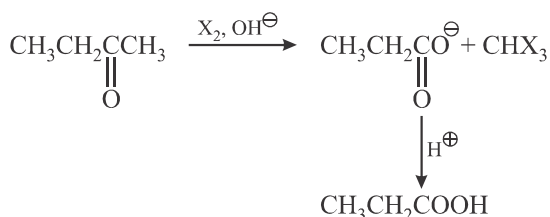
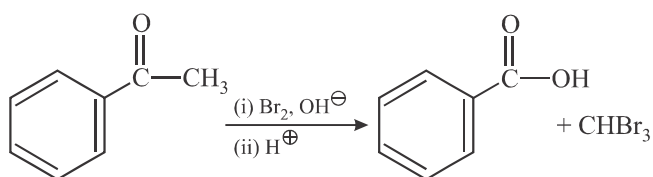
Ketones are oxidized under vigorous conditions such as hot  $\text{K}_2\text{Cr}_2\text{O}_7/\text{H}^+$  and give mixture of acids.

**Example – 17**

**Example – 18**


Keto group (with C\*) remains with smaller alkyl group.

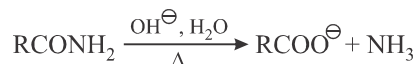
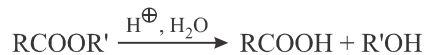
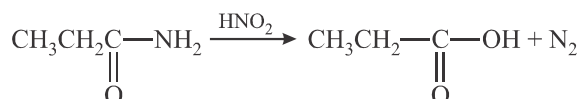
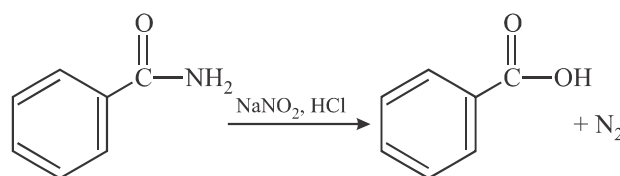
**3.6.2 Haloform Reaction**

Methyl ketone on treatment with  $\text{X}_2/\text{OH}^-$ , gives carboxylate having one carbon atom less than the original ketone – **Haloform Reaction**.

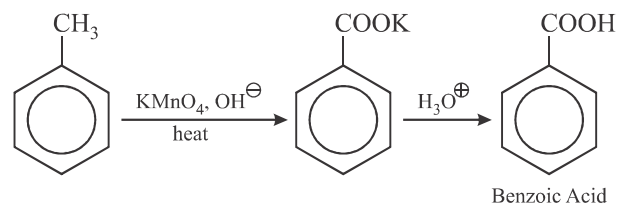
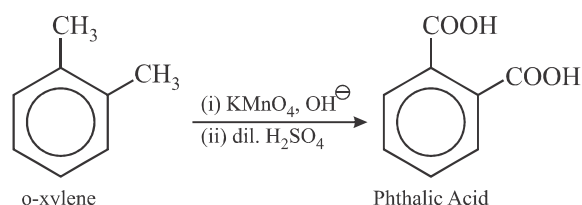
**Example – 19**

**Example – 20**

**3.7 Carboxylic Acid Derivatives**

Acyl halide, carboxylic ester, anhydride and amide all on acidic or basic hydrolysis give carboxylic acids.

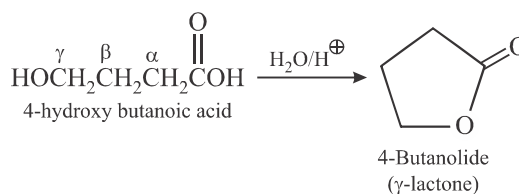
1° amide on treatment with nitrous acid.

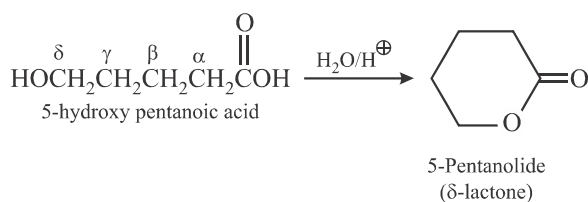

**Example – 21**

**Example – 22**

**3.8 Preparation of Aromatic Acids**
**3.8.1 Oxidation of Alkyl Benzenes**

Alkyl side chain of benzene ring can be easily oxidized to carboxylic group, irrespective of the length with acidic or alkaline  $\text{KMnO}_4$ , chromic acid or conc  $\text{HNO}_3$ .

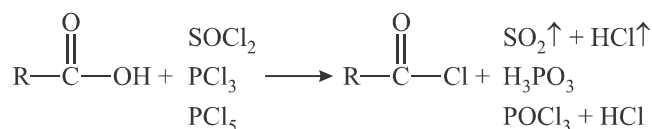
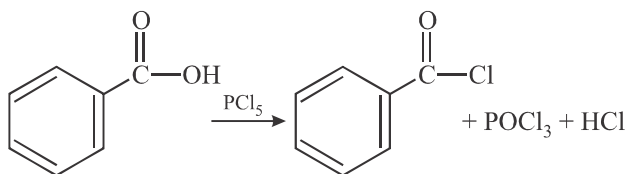
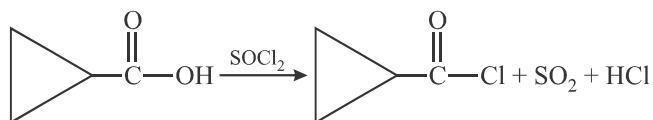
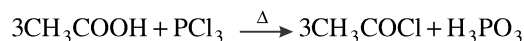
**Example – 23**

**Example – 24**


For conversion to carboxylic group, presence of  $\alpha$ -hydrogen is necessary. That is why tertiary alkyl side chains are not oxidized.

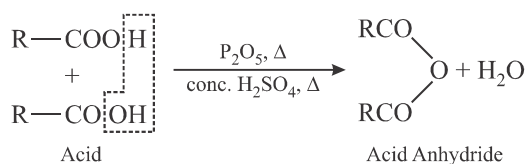
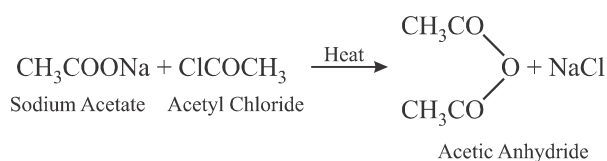
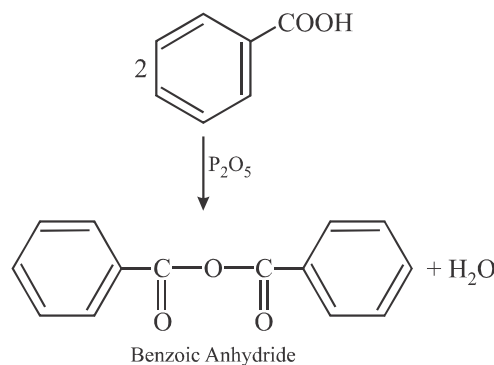


**Example – 27**


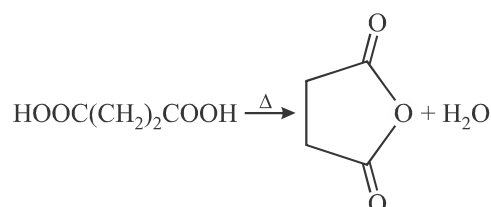
5 membered ring is called  $\gamma$ -lactone and 6 membered ring is called  $\delta$ -lactone.

**4.2.2 Formation of Acid Chlorides**

**Example – 28**


Out of these three,  $\text{SOCl}_2$  method is preferred because here other two products are gases that can easily escape.

**4.2.3 Formation of Acid Anhydrides**

**Example – 29**

**Example – 30**


Dicarboxylic acids on heating give cyclic anhydride (5 and 6 membered rings).

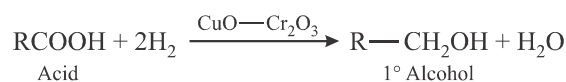
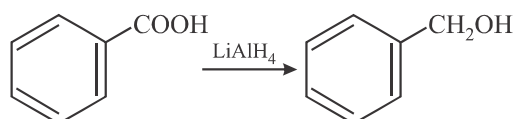
**Example – 31**


*Note...* Formic Acid ( $\text{HCOOH}$ ) does not form anhydride.

**4.2.4 Formation of Amides**

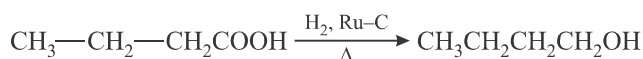
**4.3 Reduction**
**4.3.1 Reduction to Alcohols**

**Reducing Agents :**  $\text{LiAlH}_4$  in ether solution or with  $\text{H}_2$  in presence of copper chromite ( $\text{CuCr}_2\text{O}_4$ ) catalyst or  $\text{BH}_3/\text{THF}$  (or diborane,  $\text{B}_2\text{H}_6$ ) or  $\text{H}_2$ , Ru etc.


**Example – 32**


## Carboxylic Acids & Derivatives

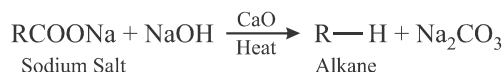
### Example – 33



## 4.4 Carboxyl Group Reactions

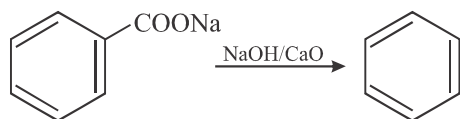
#### 4.4.1 Decarboxylation

When anhydrous alkali salt of a carboxylic acid is heated with soda lime (NaOH : CaO in ratio 3 : 1), an alkane is formed.

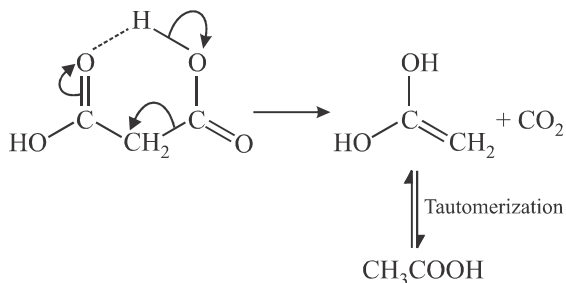
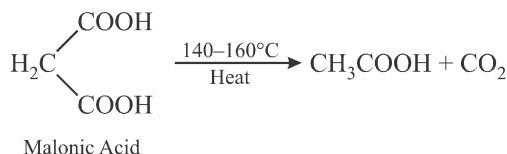


The reaction proceeds through **carbanion intermediate**. Greater the stability of carbanion, greater the rate of decarboxylation.

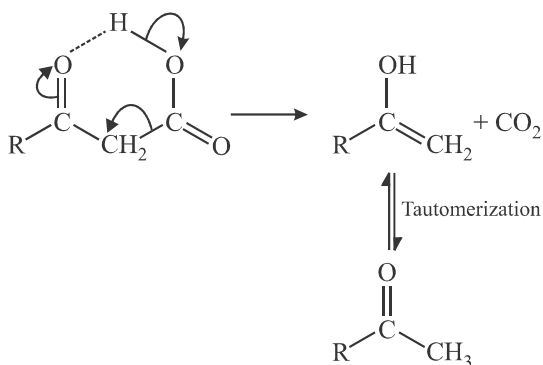
### Example – 34



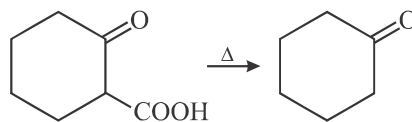
Gem dicarboxylic acids decarboxylate simply on heating.



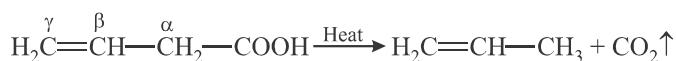
$\beta$ -keto acids undergo decarboxylation through a cyclic transition state, giving ketone.



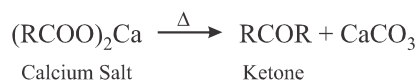
### Example – 35



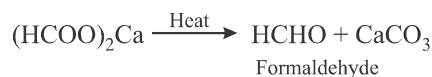
$\beta$ -unsaturated acids decarboxylate on heating giving alkenes.



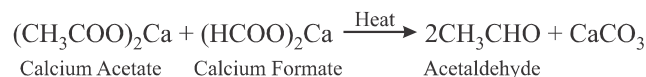
#### 4.4.2 Dry Distillation of Calcium Salts of Carboxylic Acids



When calcium formate is heated, formaldehyde is evolved.

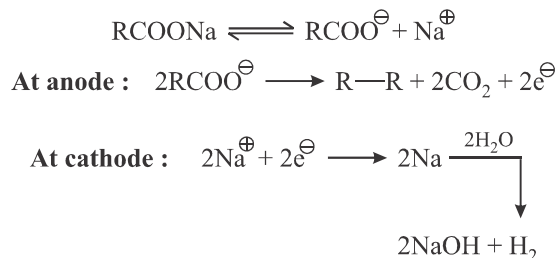


When a mixture of calcium salt of an acid (other than formic acid) and calcium formate is heated an aldehyde is formed.



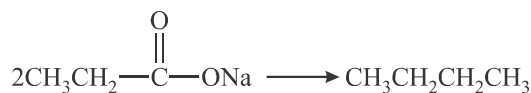
### 4.4.3 Kolbe's Synthesis

Electrolysis of sodium or potassium salts (concentrated aq. solution) of fatty acids yield alkanes.



Reaction involves radical intermediates.

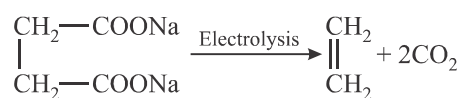
### Example – 36

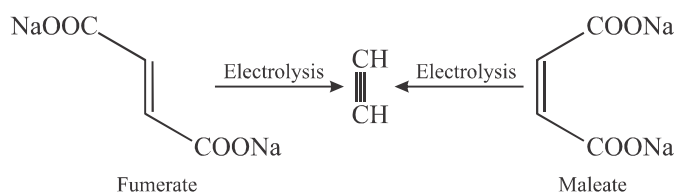
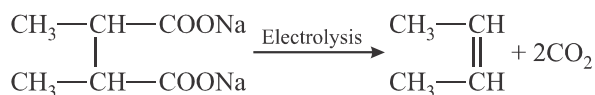


Electrolysis of salt of succinic acid and its derivative give alkene.

Electrolysis of maleic acid/fumaric acid or their salts gives corresponding alkynes.

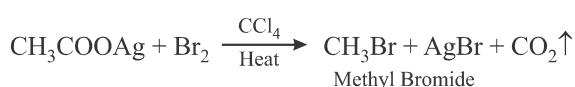
### Example – 37





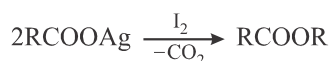
#### 4.4.4 Hunsdiecker Reaction

An alkyl halide is formed when the silver salt of monocarboxylic acid is heated with halogen. This is a step down reaction.



#### 4.4.5 Simonini Reaction

The reaction of silver salts of carboxylic acids with iodine is called Simonini Reaction.



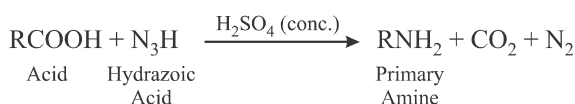
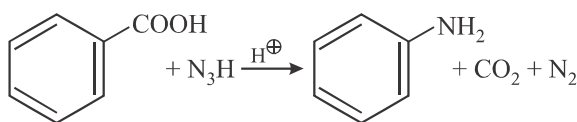
*Note...* The ratio of reagents play an important role in determination of products, namely.

- 1 : 1 ratio of salt and  $\text{I}_2$  — RI is formed
- 2 : 1 ratio of salt and  $\text{I}_2$  — RCOOR is formed
- 3 : 2 ratio of salt and  $\text{I}_2$  — Both the products are obtained

#### 4.4.6 Schmidt Reaction

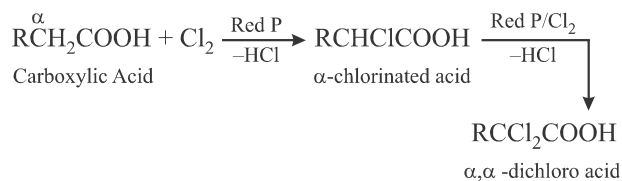
It involves treatment of hydrazoic acid with carboxylic acids. Alkyl isocyanate is the intermediate. Primary amine formed contains one carbon less than parent acid.

##### Example – 38

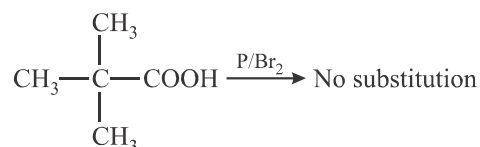
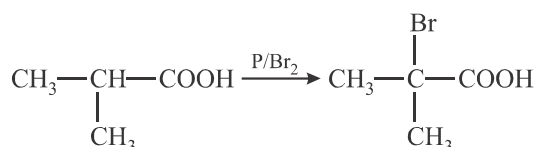
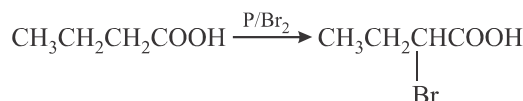


#### 4.4.7 Hell-Volhard Zelinsky Reaction

When carboxylic acid containing an  $\alpha$ -hydrogen is treated with  $\text{Cl}_2$  or  $\text{Br}_2$  in presence of a catalyst. The  $\alpha$ -hydrogen is replaced by chlorine or bromine atoms.



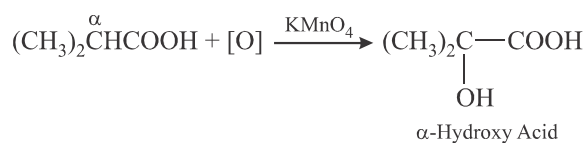
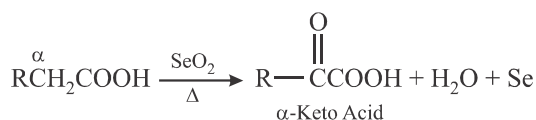
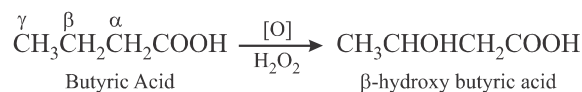
##### Example – 39



#### 4.4.8 Oxidation

Prolonged heating with powerful oxidising agents oxidises acids into  $\text{CO}_2$  and  $\text{H}_2\text{O}$ .

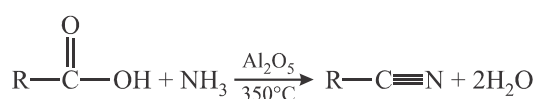
With 3%  $\text{H}_2\text{O}_2$ , they are oxidised to  $\beta$ -hydroxy acids.



#### 4.4.9 Conversion to Nitriles

When mixture of acid and  $\text{NH}_3$  are passed over heated alumina, alkane nitrile forms.

##### Example – 40



## 5. IMPORTANT CARBOXYLIC ACIDS

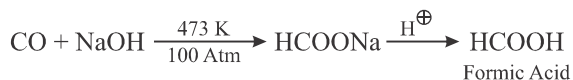
### 5.1 Formic Acid (HCOOH)

Naturally it occurs in sting of bees, wasps, red ant stinging nettles and fruits.

It was first prepared by the distillation of red ants and hence it is called as **Formic Acid** (*Latin* formica : ant)

#### 5.1.1 Preparation

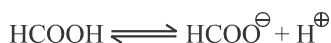
##### Industrial Preparation



Acid is regenerated by the action of  $\text{H}_2\text{SO}_4$ .

#### 5.1.2 Chemical Properties

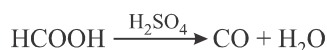
##### (A) Acidic Nature



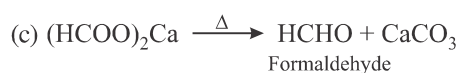
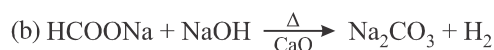
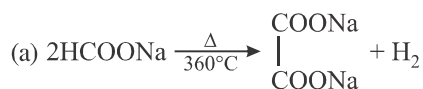
##### (B) Action of Heat



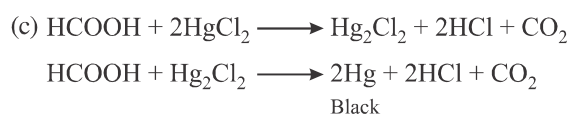
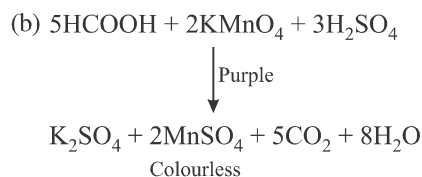
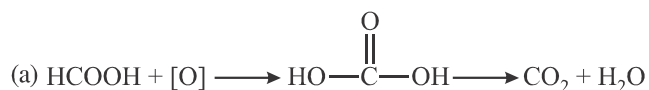
##### (C) Dehydration



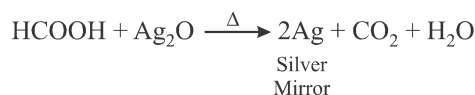
##### (D) Action of Heat on Formates



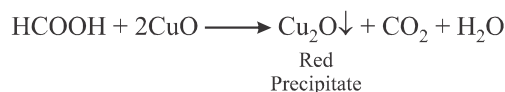
##### (E) Reducing Properties



##### (d) Tollen's Reagent



##### (e) Fehling's Solution



#### 5.1.3 Tests for HCOOH

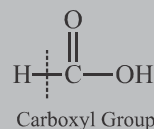
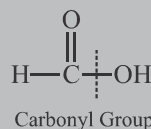
1. Its aqueous solution turns blue litmus red.
2. It forms silver mirror with Tollens' reagent.
3. It forms black ppt with  $\text{HgCl}_2$ .
4. Heated with  $\text{H}_2\text{SO}_4$ , it evolves CO which burns with blue flame on mouth of test tube.
5. On reacting with neutral solution of  $\text{FeCl}_3$ , it gives red colour which converts in brown ppt on heating.



#### 5.1.4 Uses

1. In lab preparation of CO.
2. Textile dyeing and finishing.
3. Leather tanning.
4. Coagulating agent for rubber latex.
5. As an antiseptic and treatment for gout.
6. As a strong irritant.

*Note...* 1. It is the strongest carboxylic acid.  
2. Its unique nature is due to the presence of both carboxyl and carbonyl group.

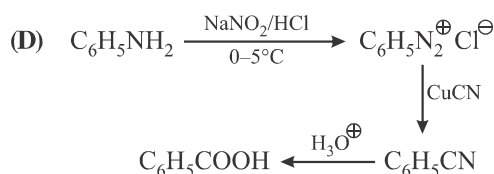
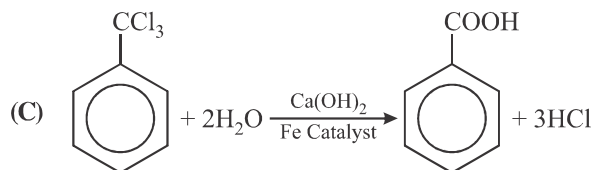
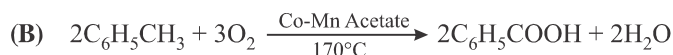
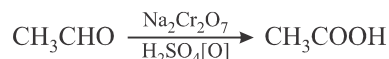
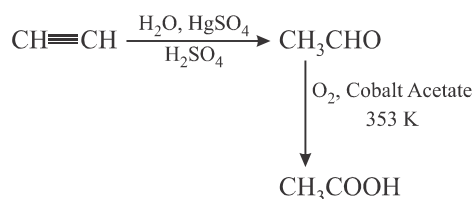
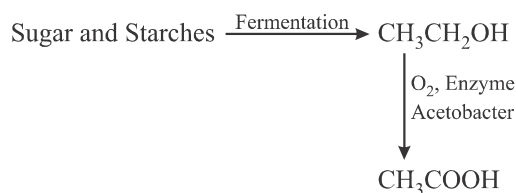
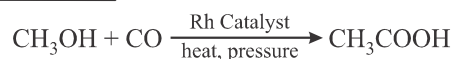


### 5.2 Acetic Acid ( $\text{CH}_3\text{COOH}$ )

In free state it is found in fermented fruit juices and in bound state it is found in biological fluids and plant extracts.

It is a by-product of alcohol industry.

### 5.2.1 Preparation



### 5.3.2 Reactions of Benzoic Acid

Benzoic acid,  $\text{C}_6\text{H}_5\text{COOH}$  shows all typical reactions of  $-\text{COOH}$  group just like aliphatic acids. The properties of the derivatives of benzoic acid are also similar to the properties of aliphatic acids.

### 5.2.2 Chemical Properties

Acetic Acid is a typical member of fatty acids. It shows all the general characteristics of mono carboxylic acids listed earlier.

### 5.2.3 Tests for $\text{CH}_3\text{COOH}$

1. Its aqueous solution turn blue litmus red.
2. Its aqueous solution gives effervescences of  $\text{CO}_2$  with  $\text{NaHCO}_3$ .
3. Neutral  $\text{CH}_3\text{COOH} \xrightarrow[\text{FeCl}_3]{\text{Neutral}}$  red colour
4. It does not affect Tollen's reagent, Fehling solution and  $\text{HgCl}_2$ .  
These reactions are used to differentiate it from  $\text{HCOOH}$ .

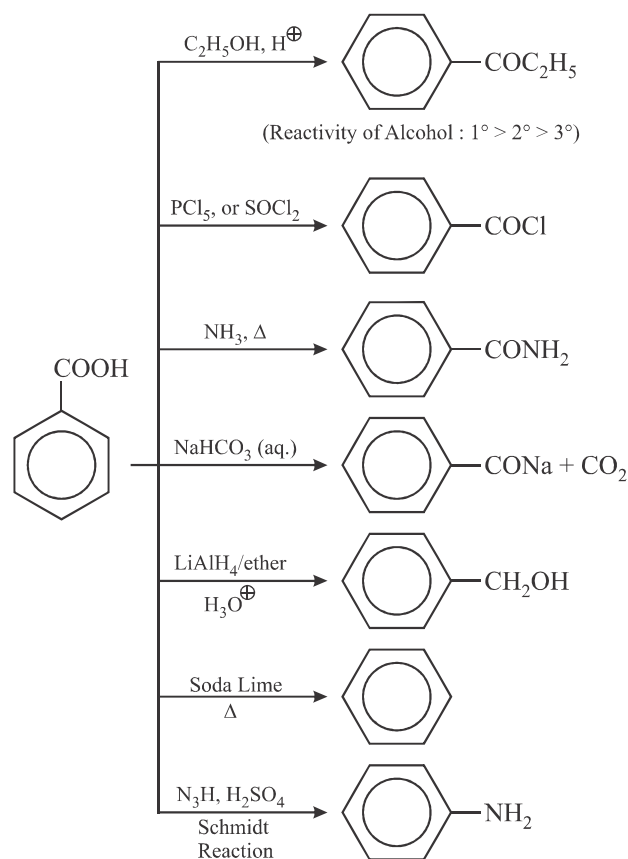
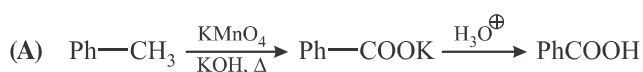
### 5.2.4 Uses

1. Good solvent and lab reagent.
2. As vinegar and table purposes in houses.
3. In manufacturing pickles.
4. Making Al, Cr, Fe acetates which are used as mordants in dyeing.

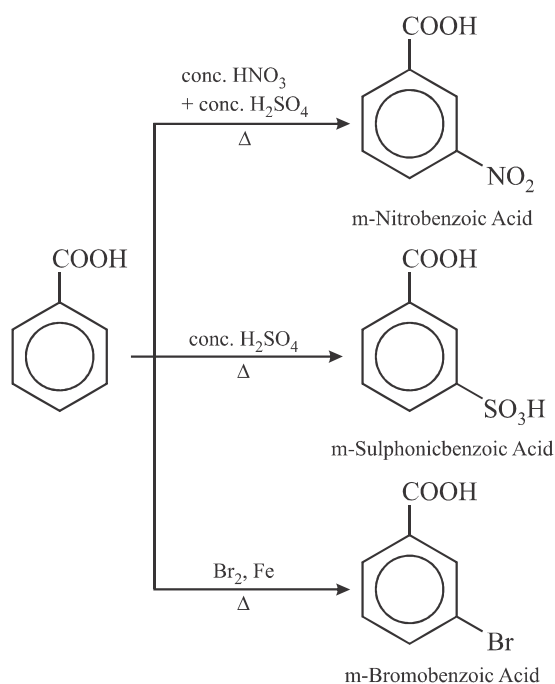
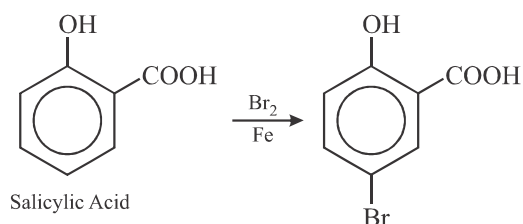
## 5.3 Benzoic Acid ( $\text{C}_6\text{H}_5\text{COOH}$ )

Naturally it is obtained from Benzoin resin. It is a benzene derivative and aromatic compound.

### 5.3.1 Preparation



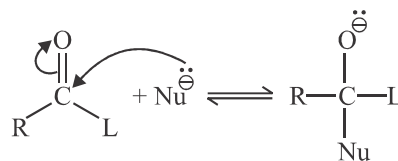
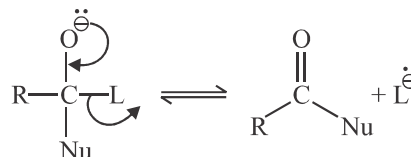
$\text{COOH}$  group attached to benzene nucleus in m-directing and a deactivating group. It also gives electrophilic substitution ( $\text{S}_\text{E}$ ) reactions.

**Example – 41****5.3.3 Uses**

1. Used as food preservative.
2. Used as precursor for many compounds.
3. Used in making medicines for fungal diseases.

**6. FUNCTIONAL DERIVATIVES OF ACIDS**

Carboxylic acids form esters, acid halides, amides and acid anhydrides as a functional derivatives. All these derivatives have a carbonyl group and yield carboxylic acids on hydrolysis. Among these derivatives, **acyl chlorides are the most reactive towards nucleophilic acyl substitution reactions and amides are least reactive.**

**General Mechanism of Nucleophilic Acyl Substitution****Step1 : Addition of Nucleophile****Step2 : Loss of Leaving Group**

**Leaving groups L :**  $\text{Cl}, \text{NH}_2, \text{RO}, \text{RCOO}$

The weaker is the base, the better is the leaving group. The basic nature of the above groups is in the order :



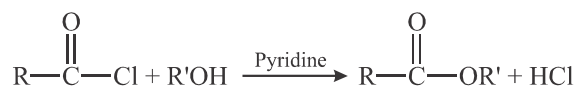
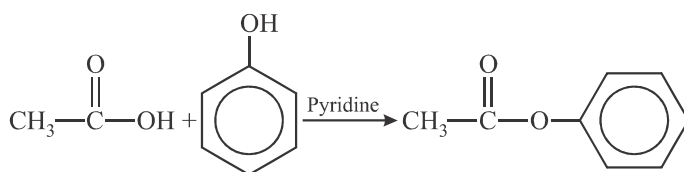
Hence, the leaving tendency of the groups is in order :

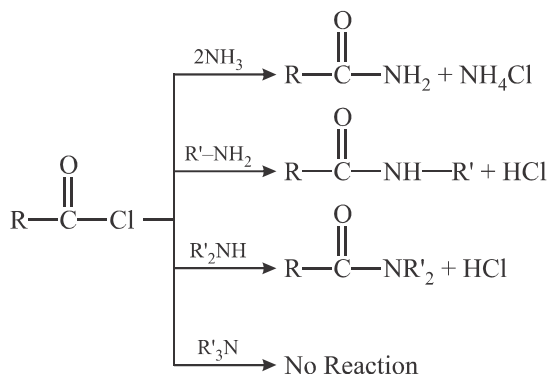
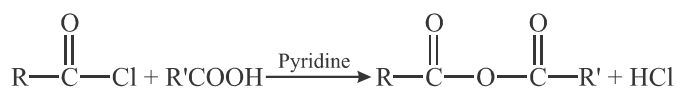
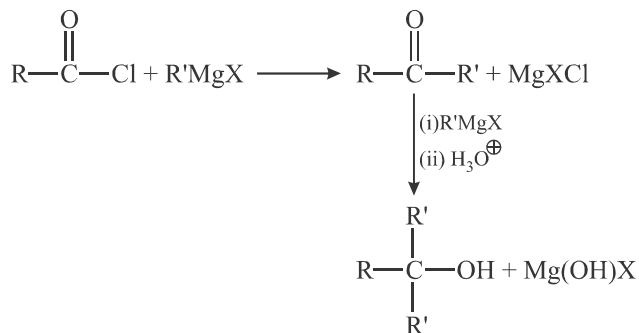
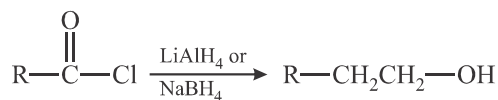
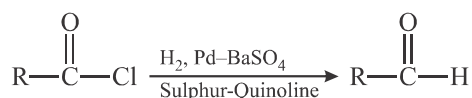
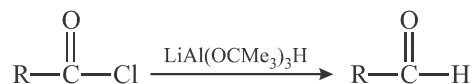
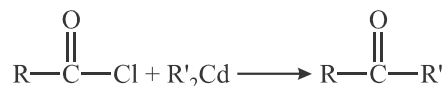


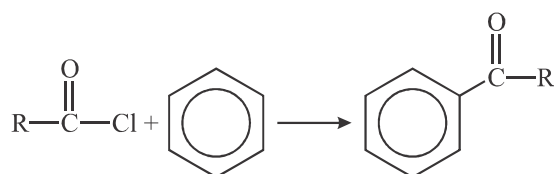
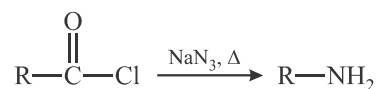
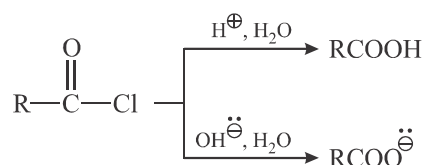
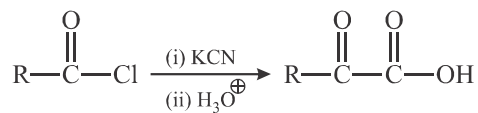
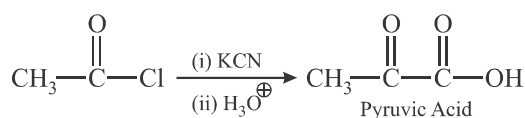
Therefore, the reactivity of the acids and their derivatives follows the order :

**6.1 Acyl Chlorides****6.1.1 Preparation**

- (A)  $\text{RCOOH} + \text{PCl}_5 \longrightarrow \text{RCOCl} + \text{POCl}_3 + \text{HCl}\uparrow$
- (B)  $\text{RCOOH} + \text{SOCl}_2 \longrightarrow \text{RCOCl} + \text{SO}_2\uparrow + \text{HCl}\uparrow$
- (C)  $\text{RCOOH} + \text{PCl}_3 \longrightarrow \text{RCOCl} + \text{H}_3\text{PO}_3$

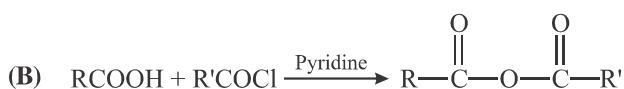
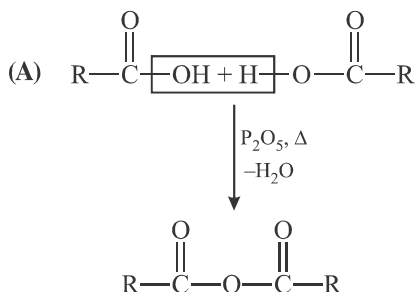
**6.1.2 Reactions****(A) Esterification****Example – 42**

**(B) Conversion to Amides**

**(C) Conversion to Anhydrides**

**(D) Conversion to Alcohols**
**(i) Grignard Reagent**

**(ii) Metal Hydrides**

**(E) Conversion to Aldehydes**
**(i) Rosenmund's Reduction**

**(ii) Metal Hydride  $\text{LiAl(OCMe}_3)_3\text{H}$** 

**(F) Conversion to Ketones**
**(i) Dialkyl Cadmium**

**(ii) Gilman's Reagent**

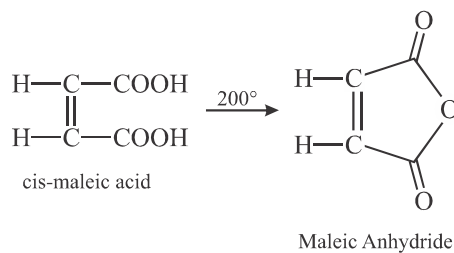
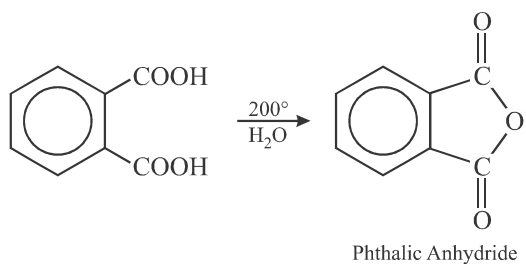
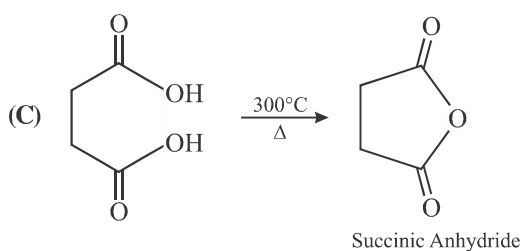
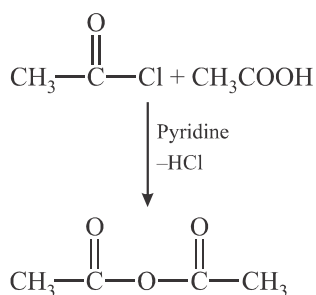
**(G) Friedel-Craft's Acylation**

**(H) Curtius Reaction**

**(I) Hydrolysis**

**(J) Formation of  $\alpha$ -Keto Acids**

**Example – 43**


## 6.2 Acid Anhydrides

### 6.2.1 Preparation



#### Example – 44

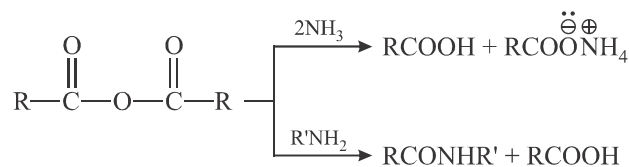


### 6.2.2 Reactions

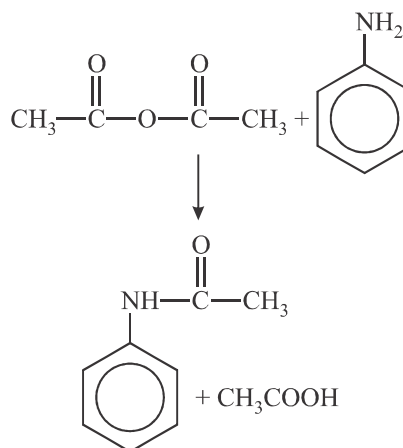
#### (A) Esterification



#### (B) Conversion to Amides



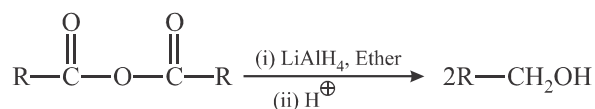
#### Example – 45



#### (C) Conversion to Acyl Chlorides

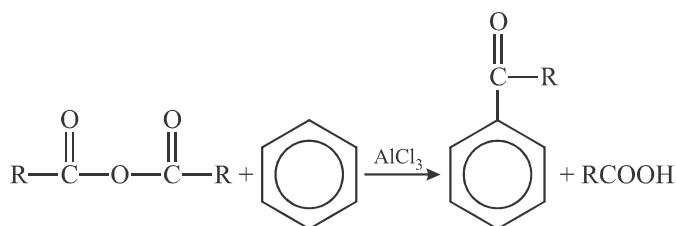


#### (D) Conversion to Alcohols

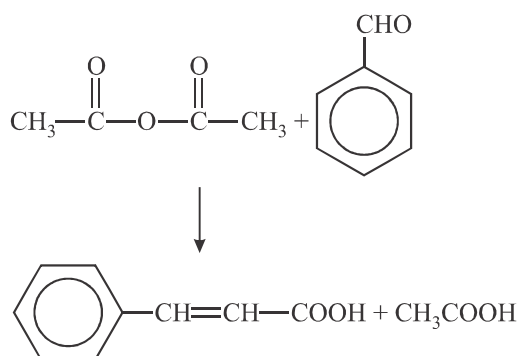


## Carboxylic Acids & Derivatives

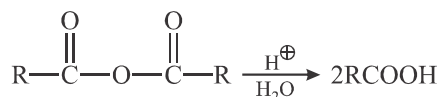
**(E) Friedel-Craft's Acylation**



### (F) Perkin Reaction



### (G) Hydrolysis

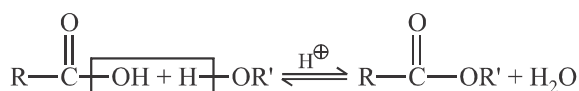


### 6.3 Esters

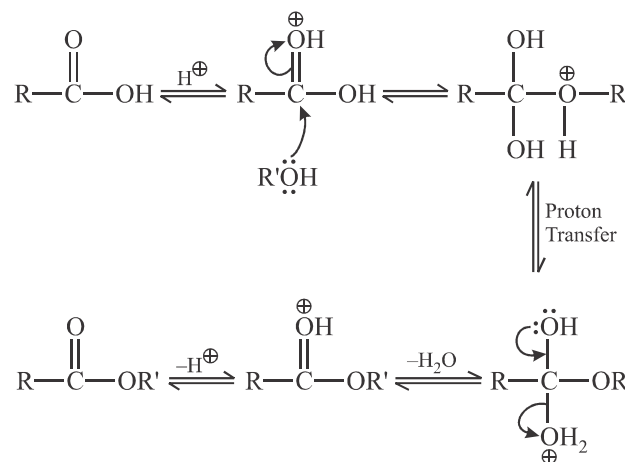
### 6.3.1 Preparation

### (A) Esterification

The esters are formed by the reaction between an acid,  $\text{RCOOH}$  and an alcohol by warming with mineral acid in a reaction called as Esterification.

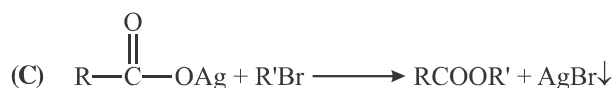
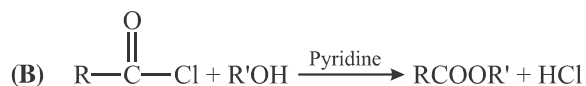


### Mechanism



Since a tetrahedral intermediate occupies more space than a planar carbonyl group, we would expect the rate of this reaction to be retarded when bulky reactants are used. When the esterification of acetic acid was compared with that of 2, 2-dimethylpropanoic acid,  $(\text{CH}_3)_3\text{C}-\text{COOH}$ , the relatively small methyl group of acetic acid is replaced by a larger *tert*-butyl group, and the bulkier acid reacted fifty times slower than acetic acid. Similarly, increasing the size of the alcohol reactant results in a similar rate reduction.

In its formation, steric factor is most important both in acids and alcohols. If acid chlorides or acid anhydrides are esterified with alcohols, it is considered as the better method.



### (E) Baeyer-Villiger Oxidation of Ketones

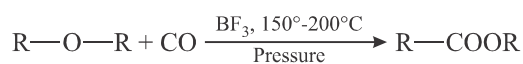
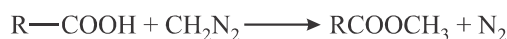


Oxygen is introduced on that side which has more electron donating alkyl group.

### Example – 46

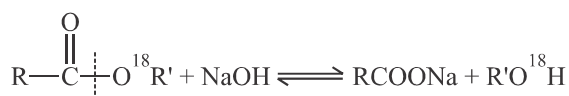
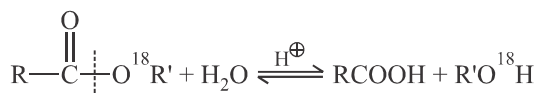


## (F) Carbonylation of Ethers


(G) Diazomethane  $CH_2N_2$ 


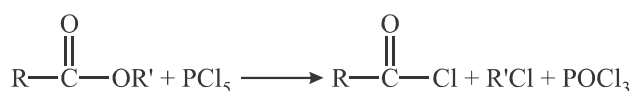
## 6.3.2 Reactions

## (A) Hydrolysis



Base catalysed hydrolysis is not reversible as carboxylate ion formed is resonance stabilised.

## (B) Conversion to Acyl Chlorides

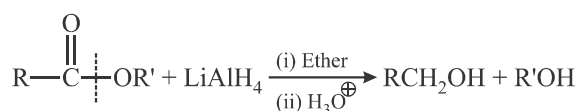


## (C) Conversion to Amides

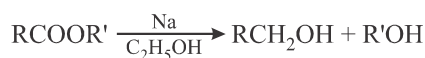


## (D) Conversion to Alcohols

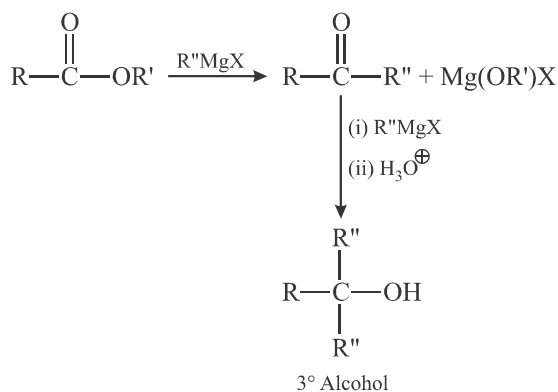
## (i) Metal Hydrides



## (ii) Bouveault-Blanc Reduction

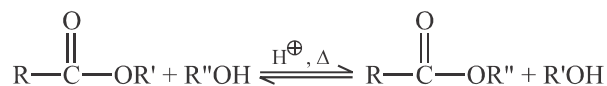


## (iii) Grignard Reagent

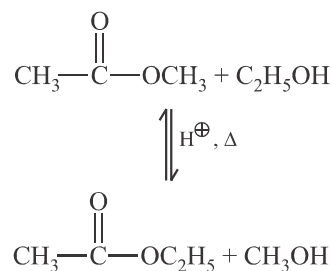


## (E) Transesterification

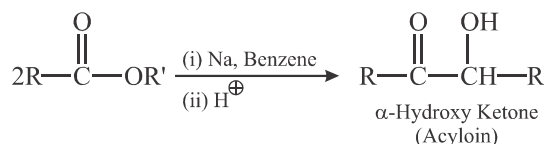
Trans esterification is the process of exchanging the alkoxy group of an ester compound by another alcohol. These reactions are often catalyzed by the addition of an acid or base.



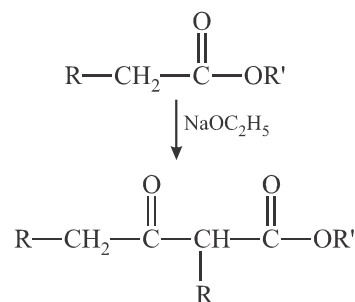
## Example – 47



## (F) Acyloin Formation

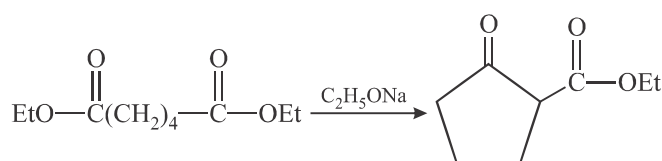


## (G) Claisen Condensation



Intramolecular claisen condensation is called as **Dieckmann Condensation**.

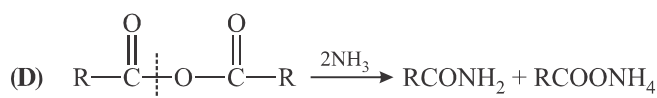
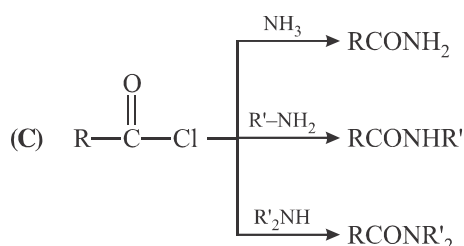
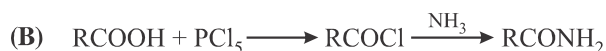
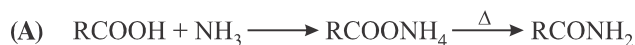
## Example – 48



### 6.4 Amides

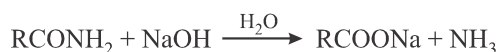
| $\text{RCONH}_2$         | $\text{RCONHR}$                     | $\text{RCONR}_2$                         |
|--------------------------|-------------------------------------|------------------------------------------|
| 1° amides<br>Alkanamides | 2° amides<br>N-Alkyl<br>Alkanamides | 3° amides<br>N, N-Dialkyl<br>Alkanamides |

#### 6.4.1 Preparation

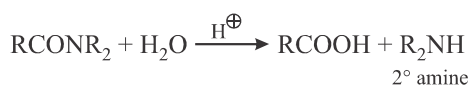
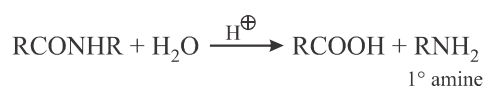


#### 6.4.2 Reactions

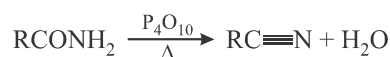
##### (A) Hydrolysis



This is typical of 1° amides to give off  $\text{NH}_3$ .



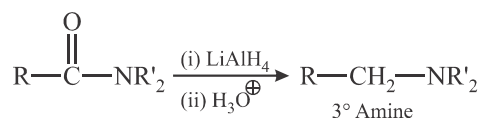
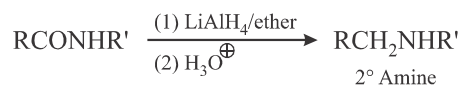
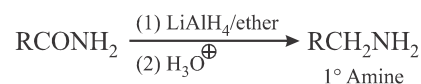
##### (B) Conversion to Cyanides



Dehydrating Reagents such as  $\text{P}_2\text{O}_5$ ,  $\text{SOCl}_2$  or  $\text{POCl}_3$  is used.

##### (C) Conversion to Amines

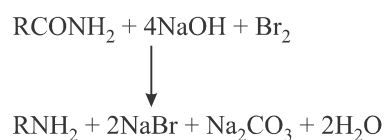
Amides are reduced to amines by reaction with reducing agents such as  $\text{LiAlH}_4$  or  $\text{Na/C}_2\text{H}_5\text{OH}$ .



##### (D) Reaction with $\text{HNO}_2$



##### (E) Hofmann Bromamide Degradation

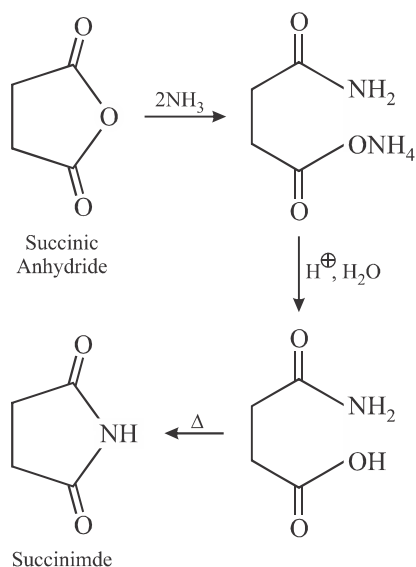


Substituted amides do not give this reaction.

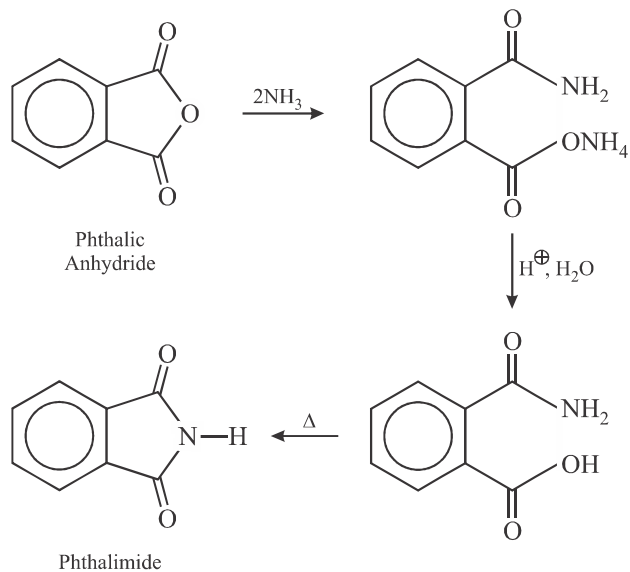
##### (F) Conversion to Imides

Cyclic anhydrides are converted first into an amide on reaction with  $\text{NH}_3$  and then heated to form **imides**.

#### Example – 49

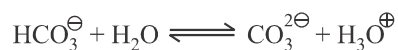
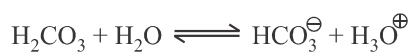
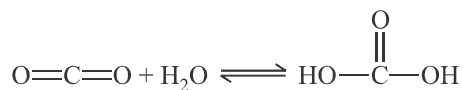


## Example – 50

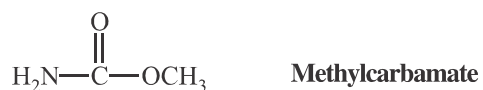
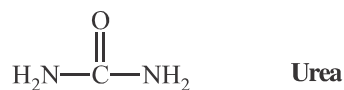
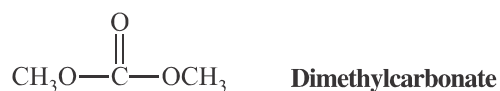


## 6.5 Carbonic Acid Derivatives

Aqueous carbonic acid ( $\text{CO}_2$  gas dissolved in  $\text{H}_2\text{O}$ ) is unstable and behaves as dibasic acid.



## Derivatives of Carbonic Acid



## EXERCISE - 1 : BASIC OBJECTIVE QUESTIONS

- The general formula for monocarboxylic acids is  
(a)  $C_n H_{2n} O$  (b)  $C_n H_{2n+1} O_2$   
(c)  $C_n H_{2n+2} O_2$  (d)  $C_n H_{2n} O_2$
- Monocarboxylic acids are functional isomers of  
(a) Alcohols (b) Ethers  
(c) Esters (d) Aldehydes and ketones.
- How many acids and esters are possible for the compound with molecular formula  $C_4 H_8 O_2$   
(a) Two acids + Two esters  
(b) Two acids + Four esters  
(c) Four acids + Two esters  
(d) Three acids + Three esters
- How many stereoisomers (geometrical and optical) are possible for the following compound ?  
 $CH_3 CH=CH CH_2 CH(OH) COOH$   
(a) 1 (b) 2  
(c) 3 (d) 4
- Which of the following has the highest boiling point ?  
(a)  $CH_3 OCH_2 CH_3$  (b)  $CH_3 CH_2 CH_2 OH$   
(c)  $CH_3 COOH$  (d)  $CH_3 CH_2 CH_2 CH_3$
- The carboxylic acid which has maximum solubility in water is  
(a) Phthalic acid (b) Succinic acid  
(c) Malonic acid (d) Salicylic acid
- The acidic nature of carboxylic acids,  
 $RCOOH \rightarrow RCOO^- + H^+$  is due to  
(a) High degree of ionization of carboxylic acids  
(b) Greater resonance stabilization of the carboxylic acid over the carboxylate ion  
(c) Greater resonance stabilization of the carboxylate ion over the carboxylic acid  
(d) Solubility of carboxylic acids in water.
- Which of the following would be expected to be the most highly ionized in water ?  
(a)  $ClCH_2 CH_2 CH_2 COOH$  (b)  $CH_3 CCl_2 CH_2 COOH$   
(c)  $CH_3 CH_2 CCl_2 COOH$  (d)  $CH_3 CH_2 CHCl COOH$
- Which of the following is a polyprotic acid  
(a) Acetic acid (b) Benzoic acid  
(c) Salicylic acid (d) Oxalic acid
- Maximum keto-enol tautomerism is shown by  
(a)  $CH_3 COOCH_3$   
(b)  $CH_3 COCH_2 CH_3$   
(c)  $CH_3 CH_2 COOCH_2 CH_3$   
(d)  $CH_3 COCH_2 COOC_2 H_5$
- The decreasing order of reactivity of  
(i)  $CH_3 COCl$  (ii)  $CH_3 COOC_2 H_5$   
(iii)  $CH_3 CONH_2$  (iv)  $(CH_3 CO)_2 O$   
Towards nucleophilic acyl substitution is  
(a) (i) > (iv) > (iii) > (ii)  
(b) (i) > (iv) > (ii) > (iii)  
(c) (iv) > (iii) > (i) > (ii)  
(d) (iii) > (i) > (iv) > (ii)
- $$\begin{array}{c} O \\ || \\ H-C-Cl \end{array}$$
 is called  
(a) Acetyl chloride (b) Formyl chloride  
(c) Chloreton (d) Oxochloromethane
- In the anion  $HCOO^-$  the two carbon-oxygen bonds are found to be of equal length. What is the reason for it.  
(a) Electronic orbitals of carbon atom are hybridised  
(b) The  $C=O$  bond is weaker than  $C-O$  bond  
(c) The anion  $HCOO^-$  has two resonating structures.  
(d) The anion is obtained by removal of a proton from the acid molecule

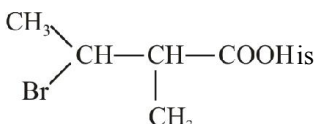
14. The IUPAC name for



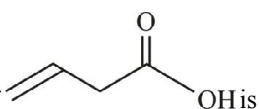
- 1-Hydroxy-4-methylpentanoic acid
- 1-Methyl-3-hydroxypentanoic acid
- 4-Hydroxy-2-methylpentanoic acid
- 4-Carboxypentanol-2.

15. The IUPAC name of  $\text{C}_6\text{H}_5\text{COCl}$  is

- chlorobenzyl ketone
- benzene chloroketone
- benzenecarbonyl chloride
- chlorophenyl ketone

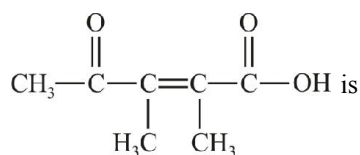
16. The IUPAC name of  is

- 2-Bromo-3-methylbutanoic acid
- 2-Methyl-3-bromobutanoic acid
- 3-Bromo-2-methylbutanoic acid
- 3-Bromo-2,3-dimethylpropanoic acid

17. The IUPAC name of  is

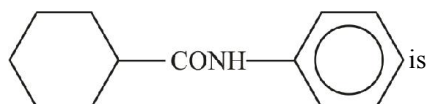
- but-3-enoic acid
- but-1-enoic acid
- pen-4-enoic acid
- prop-2-enoic acid

18. The IUPAC name of the molecule



- 4-oxo-2,3-dimethylpent-2-en-1-oic acid
- 3-carboxy-3-methylpent-2-en-3-one
- 4-carboxy-3-methylpent-3-en-2-one
- 2,3-dimethyl-4-oxopent-2-en-1-oic acid

19. The IUPAC name of



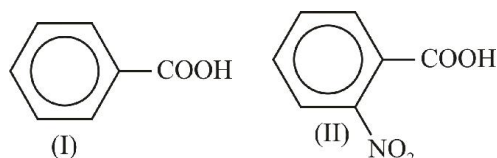
- N-Cyclohexylbenzamide
- N-Phenyl-N-cyclohexylmethanamide
- N-Phenylcyclohexanecarboxamide
- N-Cyclohexyl-N-phenylmethanamide

20. Rewrite the following in the increasing order of acidity ?

- Benzoic acid
- p-methoxybenzoic acid
- o-methoxybenzoic acid

- (i) < (ii) < (iii)
- (iii) < (i) < (ii)
- (ii) < (i) < (iii)
- (iii) < (ii) < (i)

21. Increasing value of dissociation constant  $K_a$  of



- I < II < III
- II < III < I
- III < II < I
- I < III < II

22. The increasing order of acidity of  $\alpha$ ,  $\beta$  and  $\gamma$ -chlorobutyric acids is

- $\alpha < \beta < \gamma$
- $\beta < \gamma < \alpha$
- $\gamma < \alpha < \beta$
- $\gamma < \beta < \alpha$

23. Which one of the following orders of acid strength is correct

- $\text{RCOOH} > \text{HC} \equiv \text{CH} > \text{HOH} > \text{ROH}$
- $\text{RCOOH} > \text{ROH} > \text{HOH} > \text{HC} \equiv \text{CH}$
- $\text{RCOOH} > \text{HOH} > \text{ROH} > \text{HC} \equiv \text{CH}$
- $\text{RCOOH} > \text{HOH} > \text{HC} \equiv \text{CH} > \text{ROH}$

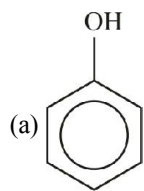
24. The weakest acid among the following is

- $\text{CH}_3\text{COOH}$
- $\text{Cl}_2\text{CHCOOH}$
- $\text{ClCH}_2\text{COOH}$
- $\text{Cl}_3\text{CCOOH}$

25. Which acid is strongest or Which is most acidic

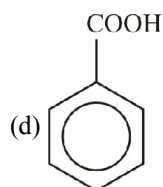
- (a)  $\text{Cl}_2\text{CHCOOH}$  (b)  $\text{ClCH}_2\text{COOH}$   
(c)  $\text{CH}_3\text{COOH}$  (d)  $\text{Cl}_3\text{CCOOH}$

26. Which of the following is the weakest acid

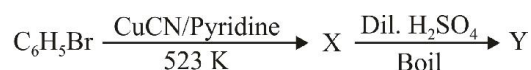


(b)  $\text{CH}_3\text{COOH}$

(c)  $\text{HCOOH}$



27. In the following reaction sequence.



The product Y is

- (a) Benzonitrile (b) Benzene  
(c) Benzoic acid (d) Benzamide.

28. Acetic acid is obtained when

- (a) Methyl alcohol is oxidised with potassium permanganate  
(b) Formaldehyde is oxidised with potassium dichromate and sulphuric acid  
(c) Acetonitrile is hydrolysed with a dilute mineral acid  
(d) Glycerol is heated with sulphuric acid.

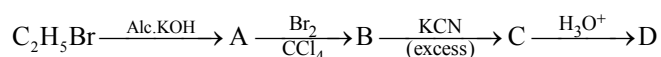
29. Ethanoic acid is prepared commercially from

- (a) Ethyne (b) Ethanol  
(c) Methanal (d) Both ethyne and ethanol.

30. Hydrolysis of hydrogen cyanide results in the formation of

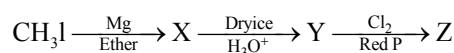
- (a) Formic acid (b) Oxalic acid  
(c) Formaldehyde (d) Acetaldehyde

31. The acid D obtained through the following sequence of reactions is



- (a) Succinic acid (b) Malonic acid  
(c) Maleic acid (d) Oxalic acid

32. Identify Z in the following reaction sequence



- (a)  $\text{CH}_3\text{COOH}$  (b)  $\text{CH}_3\text{MgI}$   
(c)  $\text{CH}_3\text{COCl}$  (d)  $\text{ClCH}_2\text{COOH}$ .

33. When benzyl alcohol is oxidised with  $\text{KMnO}_4$ , the product obtained is

- (a) Benzaldehyde (b) Benzoic acid  
(c)  $\text{CO}_2$  and  $\text{H}_2\text{O}$  (d) None of these

34. Phenol can be distinguished from acetic acid by

- (a) Blue litmus test (b) Sodium bisulphite test  
(c) With  $\text{FeCl}_3$  solution (d) Tollen's reagent test.

35. Tollen's reagent is not reduced by

- (a) Formic acid (b) Acetaldehyde  
(c) Benzaldehyde (d) Acetic acid

36. Fehling's solution can make distinction between following pairs

- I:  $\text{CH}_3\text{COOH}$ ,  $\text{HCOOH}$  II:  $\text{CH}_3\text{COOH}$ ,  $\text{HCHO}$   
III:  $\text{C}_6\text{H}_5\text{CHO}$ ,  $\text{HCHO}$  IV:  $\text{CH}_3\text{CHO}$ ,  $\text{HCHO}$

Select correct pairs :

- (a) I, II, III (b) II, III, IV  
(c) II, III (d) II only

37. Which one of the following compounds forms a red coloured solution on treatment with neutral  $\text{FeCl}_3$  solution

- (a)  $\text{CH}_3\text{COCH}_3$  (b)  $\text{CH}_3\text{OCH}_3$   
(c)  $\text{CH}_3\text{CH}_2\text{OH}$  (d)  $\text{CH}_3\text{COOH}$

38. Formic acid can reduce

- (a) Tollen's reagent (b) Mercuric chloride  
(c)  $\text{KMnO}_4$  (d) All of these

39. Which of the following will not undergo Hell - vol - hard Zelinsky (HVZ) reaction ?

- (a)  $\text{HCOOH}$  (b)  $\text{CH}_3\text{COOH}$   
(c)  $\text{CH}_3\text{CH}_2\text{COOH}$  (d)  $\text{CH}_3\text{CHBrCOOH}$

40. Propanoic acid or its sodium salt can be converted into alkanes by reduction with  $\text{HI/red P}$  or decarboxylation reaction or kolbe's reduction. Which of the following alkanes is not formed in any of these reactions ?

- (a) Methane (b) Ethane  
(c) Propane (d) Butane

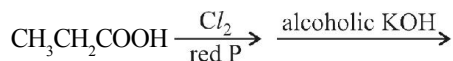
41. The reaction :  $\text{RCOOAg} + \text{Br}_2 \xrightarrow{\text{CCl}_4, \text{Reflux}}$   
 $\text{R}-\text{Br} + \text{AgBr} + \text{CO}_2$   
 is called  
 (a) Wurtz reaction (b) Hunsdiecker reaction  
 (c) Friedel-Crafts reaction  
 (d) Kolbe's reaction
42. Among the following, the acid which undergoes fastest decarboxylation is  
 (a)  $\text{Ph}-\overset{\text{O}}{\parallel}{\text{C}}-\text{COOH}$  (b)  $\text{Ph}-\overset{\text{O}}{\parallel}{\text{C}}-\text{CH}_2-\text{COOH}$   
 (c)  $\text{PhCOOH}$  (d)  $\text{Ph}-\text{CHOH}-\text{COOH}$
43. Benzoyl chloride on treatment with ammonia gives  
 (a) Benzamide (b) Acetamide  
 (c) Benzylamine (d) Benzoic acid
44. The reaction,  
 $\text{RCOOR}' + \text{R}''\text{OH} \xrightarrow{\text{H}^+ \text{ or } \text{OH}^-} \text{RCOOR}'' + \text{R}'\text{OH}$   
 is called  
 (a) Esterification (b) Trans-esterification  
 (c) Saponification (d) Hydrolysis
45. The reaction,  $\text{CH}_3\text{COOH} + \text{CH}_3\text{OH} \xrightarrow{\text{H}^+} \text{CH}_3\text{COOCH}_3 + \text{H}_2\text{O}$  is called  
 (a) Acidification reaction  
 (b) Dehydration reaction  
 (c) Dehydrogenation reaction  
 (d) Esterification reaction
46. An alkyl cyanide forms an amide when it is treated with  
 (a)  $\text{H}_2\text{O} + \text{HCl}$  (b)  $\text{NaOH} + \text{H}_2\text{O}$   
 (c)  $\text{H}_2\text{O}_2 + \text{NaOH}$  (d)  $\text{H}_2\text{SO}_4 + \text{H}_2\text{O}$
47.  $\text{CH}_3\text{COCl} \xrightarrow[\text{Pd/BaSO}_4]{\text{H}_2} \text{CH}_3\text{CHO} + \text{HCl}$ ;  
 The above reaction is called  
 (a) Reimer-Tiemann reaction  
 (b) Cannizzaro reaction  
 (c) Rosenmund reaction  
 (d) Reformatsky reaction
48. Acetic acid is weak acid than sulphuric acid because  
 (a) It decompose on increasing temperature  
 (b) It has less degree of ionisation  
 (c) It has  $-\text{COOH}$  group  
 (d) None of these
49. Heating a mixture of ethyl alcohol and acetic acid in presence of conc.  $\text{H}_2\text{SO}_4$  produces a fruity smelling compound. This reaction is called  
 (a) Neutralisation (b) Ester hydrolysis  
 (c) Esterification (d) Williamson's synthesis
50. Which reagent will bring about the conversion of carboxylic acids into esters  
 (a)  $\text{C}_2\text{H}_5\text{OH}$  (b) Dry  $\text{HCl} + \text{C}_2\text{H}_5\text{OH}$   
 (c)  $\text{LiAlH}_4$  (d)  $\text{Al}(\text{OC}_2\text{H}_5)_3$
51. Acetyl chloride cannot be obtained by treating acetic acid with  
 (a)  $\text{CHCl}_3$  (b)  $\text{SOCl}_2$   
 (c)  $\text{PCl}_3$  (d)  $\text{PCl}_5$
52.  $\text{CH}_3\text{CH}_2\text{COOH} \xrightarrow{\text{Cl}_2, \text{Red P}} \text{X} \xrightarrow{\text{Alic. KOH}} \text{Y}$   
 Compound Y is  
 (a)  $\text{CH}_3\text{CH}_2\text{OH}$  (b)  $\text{CH}_3\text{CH}_2\text{CN}$   
 (c)  $\text{CH}_2=\text{CHCOOH}$  (d)  $\text{CH}_3\text{CH}(\text{CN})\text{COOH}$
53. Acetic anhydride reacts with excess of ammonia to form  
 (a)  $2\text{CH}_3\text{COONH}_4$  (b)  $2\text{CH}_3\text{CONH}_2$   
 (c)  $\text{CH}_3\text{CONH}_2 + \text{CH}_3\text{COONH}_4$   
 (d)  $2\text{CH}_3\text{COOH}$
54. Which will not give acetamide on reaction with ammonia  
 (a) Acetic acid (b) Acetyl chloride  
 (c) Acetic anhydride (d) Methyl formate
55. In the following reaction, product P is  
 $\text{R}-\overset{\text{O}}{\parallel}{\text{C}}-\text{Cl} \xrightarrow[\text{Pd-BaSO}_4]{\text{H}_2} \text{P}$   
 (a)  $\text{RCH}_2\text{OH}$  (b)  $\text{RCOOH}$   
 (c)  $\text{RCHO}$  (d)  $\text{RCH}_3$

56. Treatment of benzoic acid with  $\text{Cl}_2/\text{FeCl}_3$  will give  
(a) *p*-chlorobenzoic acid (b) *o*-chlorobenzoic acid  
(c) 2,4-dichlorobenzoic acid (d) *m*-chlorobenzoic acid
57. Acetic anhydride is obtained from acetyl chloride by the reaction of  
(a)  $\text{P}_2\text{O}_5$  (b)  $\text{H}_2\text{SO}_4$   
(c)  $\text{CH}_3\text{COONa}$  (d)  $\text{CH}_3\text{COOH}$
58. In esterification,  $\text{OH}^\ominus$  ion for making  $\text{H}_2\text{O}$  comes from  
(a) Acid (b) Alcohol  
(c) Ketone (d) Carbohydrate
59. Hydrolysis of acetamide produces  
(a) Acetic acid (b) Acetaldehyde  
(c) Methylamine (d) Formic acid
60. When propionic acid is treated with aqueous sodium bicarbonate  $\text{CO}_2$  is liberated. The 'C' of  $\text{CO}_2$  comes from  
(a) Methyl group (b) Carboxylic acid group  
(c) Methylene group (d) Bicarbonate
61. Two moles of acetic acid are heated with  $\text{P}_2\text{O}_5$ , The product formed is  
(a) 2 moles of ethyl alcohol (b) Formic anhydride  
(c) Acetic anhydride (d) 2 moles of methyl cyanide
62. The principal organic product formed in the reaction  
$$\text{CH}_2 = \text{CH}(\text{CH}_2)_8\text{COOH} + \text{HBr} \xrightarrow{\text{peroxide}} \text{is}$$
  
(a)  $\text{CH}_3\text{CHBr}(\text{CH}_2)_8\text{COOH}$  (b)  $\text{CH}_2 = \text{CH}(\text{CH}_2)_8\text{COBr}$   
(c)  $\text{CH}_2\text{BrCH}_2(\text{CH}_2)_8\text{COOH}$   
(d)  $\text{CH}_2 = \text{CH}(\text{CH}_2)_7\text{CHBrCOOH}$
63. Ethyl acetate reacts with  $\text{CH}_3\text{MgBr}$  to form  
(a) Secondary alcohol (b) Tertiary alcohol  
(c) Primary alcohol and acid  
(d) Acid

## EXERCISE - 2 : PREVIOUS YEAR JEE MAINS QUESTION

1. End product of the following reaction is

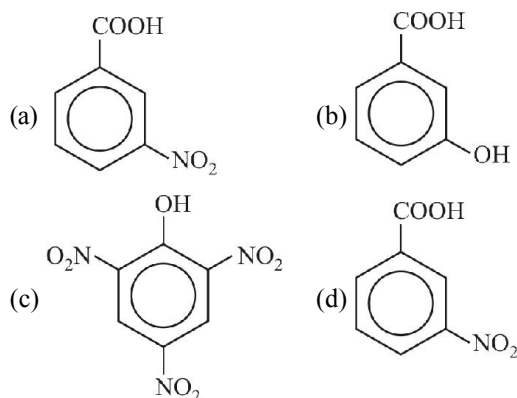
(2002) 6.



- (a)  $\text{CH}_3\text{CH}(\text{OH})\text{COOH}$  (b)  $\text{CH}_2\text{CH}_2\text{COOH}$   
(c)  $\text{CH}_2=\text{CH}-\text{COOH}$  (d)  $\text{CH}_2\text{CH}(\text{Cl})\text{COOH}$

2. Picric acid is

(2002)



3. Ethyl isocyanide on hydrolysis in acidic medium generates

- (a) ethylamine salt and methanoic acid (2003)  
(b) propanoic acid and ammonium salt  
(c) ethanoic acid and ammonium salt  
(d) methylamine salt and ethanoic acid

4. The general formula
- $\text{C}_n\text{H}_{2n}\text{O}_2$
- could be for open chain

- (a) diketones (b) carboxylic acids (2003)  
(c) diols (d) dialdehydes

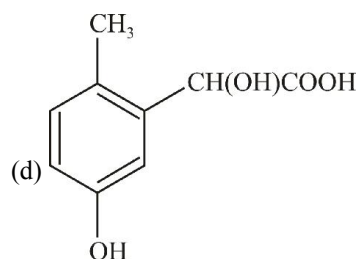
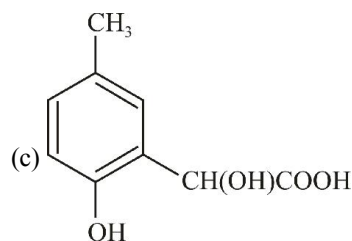
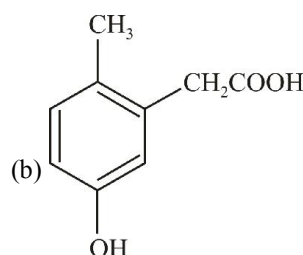
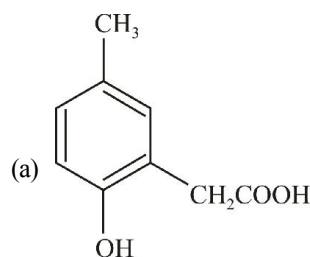
5. Consider the acidity of the carboxylic acids : (2004)

- (i)  $\text{C}_6\text{H}_4\text{COOH}$  (ii)  $\text{o-NO}_2\text{C}_6\text{H}_4\text{COOH}$   
(iii)  $\text{p-NO}_2\text{C}_6\text{H}_4\text{COOH}$  (iv)  $\text{m-NO}_2\text{C}_6\text{H}_4\text{COOH}$

Which of the following order is correct ?

- (a) (i) > (ii) > (iii) > (iv) (b) (ii) > (iv) > (iii) > (i)  
(c) (ii) > (iv) > (i) > (iii) (d) (ii) > (iii) > (iv) > (i)

p-cresol reacts with chloroform in alkaline medium to give the compound A which adds hydrogen cyanide to form the compound B. The latter, on acidic hydrolysis gives chiral carboxylic acid. The structure of the carboxylic acid is (2005)



7. The compound formed as a result of oxidation of ethyl benzene by
- $\text{KMnO}_4$
- is (2007)

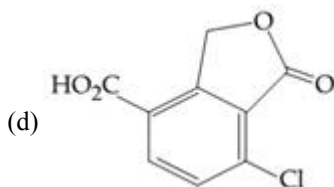
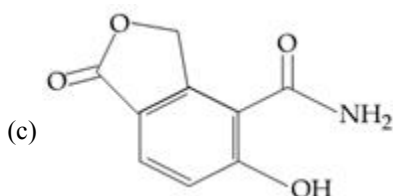
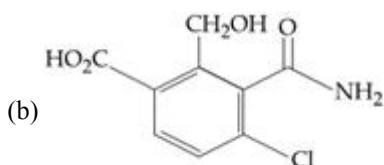
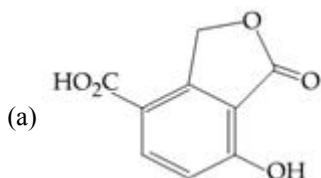
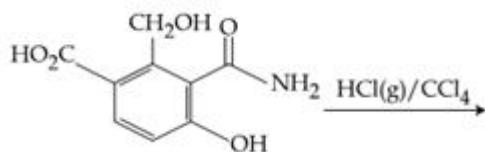
- (a) benzophenone (b) acetophenone  
(c) benzoic acid (d) benzyl alcohol

8. The major product obtained on interaction of phenol with sodium hydroxide and carbon dioxide is (2008)
- (a) benzoic acid (b) salicylaldehyde  
(c) salicylic acid (d) phthalic acid
9. Sodium ethoxide has reacted with ethanoyl chloride. The compound that is produced in the above reaction is (2010)
- (a) diethyle ether (b) 2-butanone  
(c) ethyl chloride (d) ethyl ethanoate
10. The strongest acid amongst the following compounds is (2012)
- (a)  $\text{CH}_3\text{COOH}$  (b)  $\text{HCOOH}$   
(c)  $\text{CH}_3\text{CH}_2\text{CH}(\text{Cl})\text{CO}_2\text{H}$  (d)  $\text{ClCH}_2\text{CH}_2\text{CH}_2\text{COOH}$
11. When  $\text{CH}_2=\text{CH}-\text{COOH}$  is reduced with  $\text{LiAlH}_4$ , the compound obtained will be (2013)
- (a)  $\text{CH}_3-\text{CH}_2-\text{COOH}$  (b)  $\text{CH}_2=\text{CH}-\text{CH}_2\text{OH}$   
(c)  $\text{CH}_3-\text{CH}_2-\text{CHO}$   
(d)  $\text{CH}_3-\text{CH}_2-\text{CH}_2-\text{OH}$
12. In the reaction, (2014)
- $$\text{CH}_3\text{COOH} \xrightarrow{\text{LiAlH}_4} \text{A} \xrightarrow{\text{PCl}_5} \text{B} \xrightarrow{\text{Alc.KOH}} \text{C},$$
- the product C is :
- (a) Acetylene (b) Ethylene  
(c) Acetyl chloride (d) Acetaldehyde
13. In the following sequence of reactions : (2015)
- $$\text{Toluene} \xrightarrow{\text{KMnO}_4} \text{A} \xrightarrow{\text{SOCl}_2} \text{B} \xrightarrow[\text{BaSO}_4]{\text{H}_2/\text{Pd}} \text{C},$$
- the product C is :
- (a)  $\text{C}_6\text{H}_5\text{CH}_2\text{OH}$  (b)  $\text{C}_6\text{H}_5\text{CHO}$   
(c)  $\text{C}_6\text{H}_5\text{COOH}$  (d)  $\text{C}_6\text{H}_5\text{CH}_3$
14. In the Hofmann bromamide degradation reaction, the number of moles of  $\text{NaOH}$  and  $\text{Br}_2$  used per mole of amine produced are : (2016)
- (a) Four moles of  $\text{NaOH}$  and two moles of  $\text{Br}_2$ .  
(b) Two moles of  $\text{NaOH}$  and two moles of  $\text{Br}_2$ .  
(c) Four moles of  $\text{NaOH}$  and one mole of  $\text{Br}_2$ .  
(d) One mole of  $\text{NaOH}$  and one mole of  $\text{Br}_2$ .
15. Sodium salt of an organic acid 'X' produces effervescence with conc.  $\text{H}_2\text{SO}_4$ . 'X' reacts with the acidified aqueous  $\text{CaCl}_2$  solution to give a white precipitate which decolourises acidic solution of  $\text{KMnO}_4$ . 'X' is (2017)
- (a)  $\text{C}_6\text{H}_5\text{COONa}$  (b)  $\text{HCOONa}$   
(c)  $\text{CH}_3\text{COONa}$  (d)  $\text{Na}_2\text{C}_2\text{O}_4$

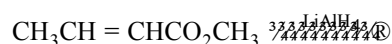
## JEE MAINS ONLINE QUESTION

1. Among the following organic acids, the acid present in rancid butter is : Online 2014 SET (1)
- (a) Butyric acid (b) Acetic acid  
(c) Lactic acid (d) Pyruvic acid
2. Bouveault-Blanc reduction reaction involves : Online 2016 SET (1)
- (a) Reduction of an acyl halide with  $\text{H}_2/\text{Pd}$ .  
(b) Reduction of an ester with  $\text{Na}/\text{C}_2\text{H}_5\text{OH}$ .  
(c) Reduction of a carbonyl compound with  $\text{Na}/\text{Hg}$  and  $\text{HCl}$ .  
(d) Reduction of an anhydride with  $\text{LiAlH}_4$ .

3. The major product expected from the following reaction is :  
Online 2017 SET (1)



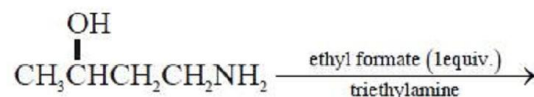
4. The major product of the following reaction is :



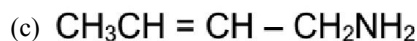
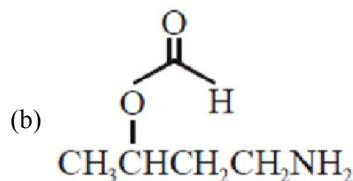
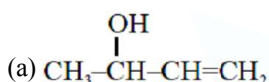
Online 2018 SET (2)

- (a)  $\text{CH}_3\text{CH}_2\text{CH}_2\text{CHO}$  (b)  $\text{CH}_3\text{CH}=\text{CHCH}_2\text{OH}$   
(c)  $\text{CH}_3\text{CH}_2\text{CH}_2\text{CO}_2\text{CH}_3$  (d)  $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{OH}$

5. The major product of the following reaction is



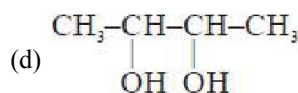
Online 2019 SET (1)



6. But-2-ene on reaction with alkaline  $\text{KMnO}_4$  at elevated temperature followed by acidification will give :

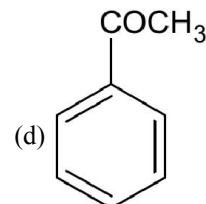
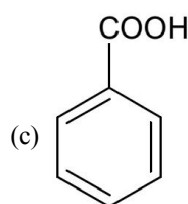
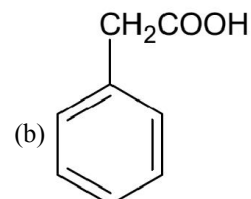
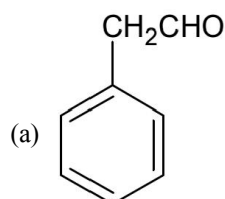
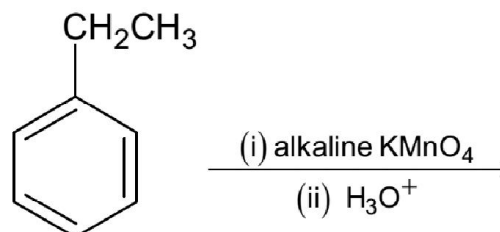
Online 2019 SET (2)

- (a) One molecule of  $\text{CH}_3\text{CHO}$  and one molecule of  $\text{CH}_3\text{COOH}$   
(b) 2 molecules of  $\text{CH}_3\text{CHO}$   
(c) 2 molecules of  $\text{CH}_3\text{COOH}$



7. The major product of the following reaction is :

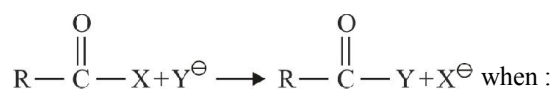
Online 2019 SET (3)



# EXERCISE - 3 : ADVANCED OBJECTIVE QUESTIONS

1. (S) A tribasic acid is  
(a) Oxalic acid (b) Tartaric acid  
(c) Lactic acid (d) Citric acid
2. (S) What is the main reason for the fact that carboxylic acids can undergo ionization  
(a) Absence of  $\alpha$ -hydrogen  
(b) Resonance stabilization of the carboxylate ion  
(c) High reactivity of  $\alpha$ -hydrogen  
(d) Hydrogen bonding
3. (S)  $\text{HCOOH}$  shows an tests of aldehyde because  
(a) It has one aldehyde group  
(b) It is member of aldehyde  
(c) All acids show tests of aldehyde  
(d) Does not show any test
4. (S) Which is more basic oxygen in an ester
- $$\text{R}-\overset{\beta}{\underset{\alpha}{\text{C}}}(\text{O})-\text{O}-\text{R}'$$
- (a) O denoted by  $\alpha$  (b) O denoted by  $\beta$   
(c) Both equally  
(d) None of the oxygen atoms is basic
5. (S)  $\text{CH}_3-\overset{\text{O}}{\parallel}{\text{C}}-\text{Cl} + \text{Nu}^\ominus \rightarrow \text{CH}_3-\overset{\text{O}}{\parallel}{\text{C}}-\text{Nu} + \text{Cl}^\ominus$   
Reactivity order of different nucleophiles  
 $(\text{NH}_2^\ominus, \text{CH}_3\text{COO}^\ominus, \text{OH}^\ominus)$  is in order  
(a)  $\text{NH}_2^\ominus < \text{CH}_3\text{COO}^\ominus < \text{OH}^\ominus$   
(b)  $\text{CH}_3\text{COO}^\ominus < \text{OH}^\ominus < \text{NH}_2^\ominus$   
(c)  $\text{NH}_2^\ominus < \text{OH}^\ominus < \text{CH}_3\text{COO}^\ominus$   
(d)  $\text{CH}_3\text{COO}^\ominus < \text{NH}_2^\ominus < \text{OH}^\ominus$
6. (S) Which of the following para-substituted benzoic acids is the most acidic  
(a)  $\text{Cl}-\text{C}_6\text{H}_4\text{COOH}$  (b)  $\text{NO}_2-\text{C}_6\text{H}_4\text{COOH}$   
(c)  $\text{HO}-\text{C}_6\text{H}_4-\text{COOH}$   
(d)  $\text{CH}_3-\text{C}_6\text{H}_4-\text{COOH}$
7. (S) Which is hydrolysed to a maximum extent ?  
(a)  $\text{CH}_3\overset{\text{O}}{\parallel}{\text{C}}\text{Cl}$  (b)  $\text{CH}_3\overset{\text{O}}{\parallel}{\text{C}}\text{NH}_2$   
(c)  $\text{CH}_3\overset{\text{O}}{\parallel}{\text{C}}\text{OC}_2\text{H}_5$  (d)  $\text{CH}_3\overset{\text{O}}{\parallel}{\text{C}}\text{CH}_3$
8. (S) Which has maximum value of equilibrium constant ?  
(a)  $\text{CH}_3\text{COOH} + \text{H}_2\text{O} \rightleftharpoons \text{CH}_3\text{COO}^- + \text{H}_3\text{O}^+$   
(b)  $\text{HCOOH} + \text{H}_2\text{O} \rightleftharpoons \text{HCOO}^- + \text{H}_3\text{O}^+$   
(c)  $\text{C}_6\text{H}_5\text{COOH} + \text{H}_2\text{O} \rightleftharpoons \text{C}_6\text{H}_5\text{COO}^- + \text{H}_3\text{O}^+$   
(d)  $\text{CF}_3\text{COOH} + \text{H}_2\text{O} \rightleftharpoons \text{CF}_3\text{COO}^- + \text{H}_3\text{O}^+$
9. (S) Select correct statement (s) :  
(a) Carboxylic acids have higher boiling points than isomeric esters because of dimer formation by intermolecular H-bonding.  
(b)  $\text{R}-\overset{\text{O}}{\parallel}{\text{C}}\text{NH}_2$  or  $\text{R}-\overset{\text{O}}{\parallel}{\text{C}}\text{NHR}$  molecules are associated by H-bonding  
(c) The presence of carbonyl group in acid derivative makes them polar compounds  
(d) All of the above statements are correct
10. (S) Resonance stabilisation is maximum in :  
(a)  $\text{R}-\overset{\text{O}}{\parallel}{\text{C}}\text{Cl}$  (b)  $\text{R}-\overset{\text{O}}{\parallel}{\text{C}}\text{OCH}_3$   
(c)  $\text{R}-\overset{\text{O}}{\parallel}{\text{C}}\text{OH}$  (d)  $\text{R}-\overset{\text{O}}{\parallel}{\text{C}}\text{NH}_2$

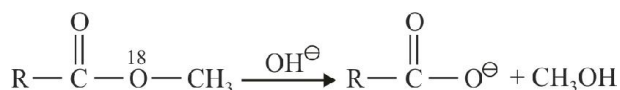
11. (S) Acyl  $S_N$  reaction is not possible



(a) X is  $NH_2$  and Y is Cl (b) X is OH and Y is  $NH_2$

(c) X is Cl and Y is OH (d) X is OR and Y is  $NH_2$

12. (S) Select correct statements for the following base-catalysed reaction



(a) It is a straight forward nucleophilic acyl substitution (addition and then ejection)

(b)  $O^{18}$  appears in methanol  $CH_3OH$

(c) The reaction is not reversible

(d) All are correct statements

13. (S) For an alkane nitrile :

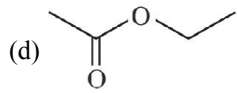
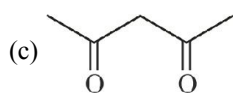
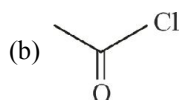
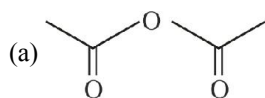
(a) nitrile carbon is  $sp$ -hybridised

(b) there can be nucleophilic attack and electrophilic attack on nitrile carbon and nitrogen respectively

(c) basic character is less than ammonia or amines

(d) all of the above statements are true

14. (S) Which is hydrolysed to the maximum extent ?



15. (A) **Assertion :** Resonance energy of carboxylate ion is much greater than that of undissociated carboxylic acid.

**Reason :** Carboxylate ion is a resonance hybrid of equivalent resonating structures.

(a) A

(b) B

(c) C

(d) D

16. (A) **Assertion :** The second dissociation constant of maleic acid is greater than fumaric acid.

**Reason :** Higher the dissociation constant of acid more is acidic character.

(a) A

(b) B

(c) C

(d) D

17. (A) **Assertion :** Carboxylic acids  $\left[ R-\overset{\overset{O}{\parallel}}{C}-OH \right]$  have a carbonyl group but it does not give the test of carbonyl group.

**Reason :** Due to resonance, the double bond character

of  $\left[ \overset{\overset{O}{\parallel}}{C} \right]$  group is greatly reduced.

(a) A

(b) B

(c) C

(d) D

18. (A) **Assertion :** Formic acid is a weaker acid than acetic acid.

**Reason :**  $pK_a$  of formic acid is less than acetic acid.

(a) A

(b) B

(c) C

(d) D

19. (A) **Assertion :** The  $pK_a$  value of acetic acid is lower than that of phenol.

**Reason :** Phenoxide ion is more resonance stabilized.

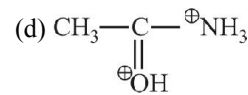
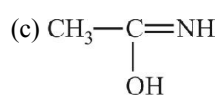
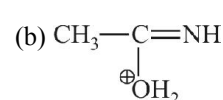
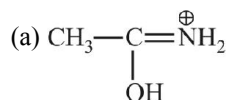
(a) A

(b) B

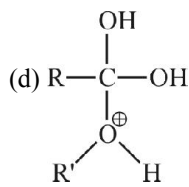
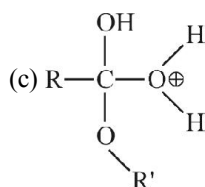
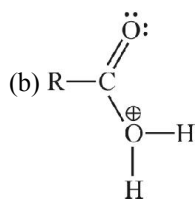
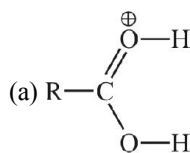
(c) C

(d) D

20. (M) Which of the following intermediate are formed during acidic hydrolysis of  $MeC \equiv N$  to  $MeCOOH$  ?



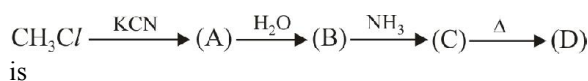
21. (M) Which of the following intermediates are involved in the acid catalysed esterification of carboxylic acid ?



22. (S) Hydrolysis of an ester gives acid *A* and alcohol *B*. *A* reduces Fehling solution and oxidation of *B* gives *A*. The ester is

- (a) Methyl formate (b) Ethyl formate  
(c) Methyl acetate (d) Ethyl acetate

23. (S) The product *D* of the reaction

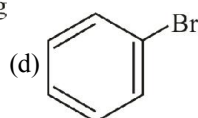
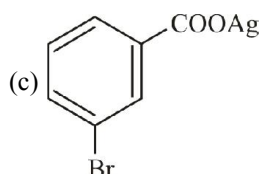
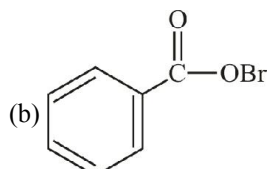
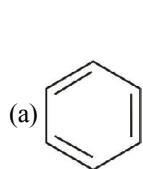


- (a)  $\text{CH}_3\text{CH}_2\text{NH}_2$  (b)  $\text{CH}_3\text{CN}$   
(c)  $\text{HCONH}_2$  (d)  $\text{CH}_3\text{CONH}_2$

24. (S) Benzoyl chloride is prepared from benzoic acid by

- (a)  $\text{Cl}_2$ , hv (b)  $\text{SO}_2\text{Cl}_2$   
(c)  $\text{SOCl}_2$  (d)  $\text{Cl}_2$ ,  $\text{H}_2\text{O}$

25. (S) Silver benzoate reacts with bromine to form



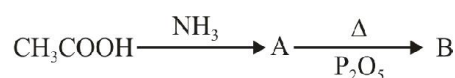
26. (S) Sulphonation of benzoic acid produces mainly

- (a) *o*-sulphobenzoic acid  
(b) *m*-sulphobenzoic acid  
(c) *p*-sulphobenzoic acid  
(d) *o*- and *p*-sulphobenzoic acid

27. (S) Which of the following undergoes hydrolysis when dissolved in water

- (a)  $\text{CH}_3\text{COONa}$  (b)  $\text{CH}_3\text{CONH}_2$   
(c) Both (a) & (b) (d)  $\text{C}_6\text{H}_5\text{CH}_3$

28. (S) Name the end product in the following series of reactions



- (a)  $\text{CH}_4$  (b)  $\text{CH}_3\text{OH}$   
(c) Acetonitrile (d) Ammonium acetate

29. (S) Oxidation of toluene with  $\text{CrO}_3$  in the presence of  $(\text{CH}_3\text{CO})_2\text{O}$  gives a product '*A*' which on treatment with aqueous  $\text{NaOH}$  produces

- (a)  $\text{C}_6\text{H}_5\text{CHO}$  (b)  $(\text{C}_6\text{H}_5\text{CO})_2\text{O}$   
(c)  $\text{C}_6\text{H}_5\text{COONa}$  (d) 2,4-diacetyl toluene

30. (S) Benzoic acid gives benzene on being heated with *X* and phenol gives benzene on being heated with *Y*. Therefore *X* and *Y* are respectively

- (a) Sodalime and copper  
(b) *Zn* dust and  $\text{NaOH}$   
(c) *Zn* dust and sodalime  
(d) Sodalime and zinc dust

31. (S) Which of the following compounds will react with  $\text{NaHCO}_3$  solution to give sodium salt and carbon dioxide

- (a) Phenol (b) *n*-hexanol  
(c) Acetic acid (d) Both (a) and (b)

32. (S) Reaction of ethyl formate with excess of  $\text{CH}_3\text{MgI}$  followed by hydrolysis gives

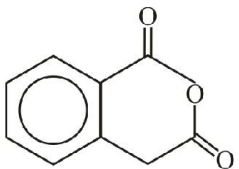
- (a) *n*-propyl alcohol (b) Ethanal  
(c) Propanal (d) Isopropyl alcohol

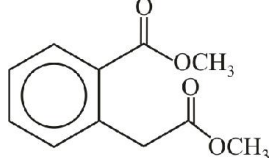
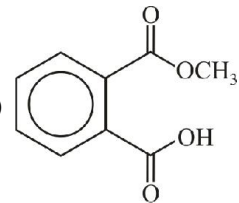
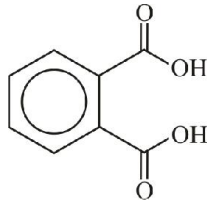
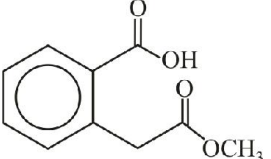
33. (S) Which of the following substances when boiled with caustic soda solution will evolve ammonia

- (a) Ethylamine (b) Aniline  
(c) Acetamide (d) Acetoxime

34. (S) An optically active compound 'X' has molecular formula  $C_4H_8O_3$ . It evolves  $CO_2$  with aq.  $NaHCO_3$ . 'X' reacts with  $LiAlH_4$  to give an achiral compound. 'X' is

- (a)  $CH_3CH_2CH(OH)COOH$   
(b)  $CH_3CH(OMe)COOH$   
(c)  $CH_3CH(CH_2OH)COOH$  (d)  $CH_3CH(OH)CH_2COOH$

35. (S)  is

- (a)   
(b)   
(c)   
(d) 

36. (S) Peroxyacetic acid ( $CH_3CO_3H$ ) is a weaker acid than acetic acid ( $CH_3CO_2H$ ) since

- (a) Negative charge in  $CH_3C(=O)OO^-$  can't be delocalised into the carbonyl group  
(b)  $CH_3$  group in  $CH_3CO_3H$  shows I-effect  
(c) Both are correct  
(d) None is correct

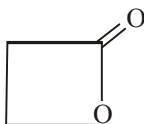
37. (S)  $CH_3CO-CH(CH_3)-CH_2CH_3$  (ester) has chiral carbon (\*).

Alkaline hydrolysis gives an acid salt and an alcohol. Select correct statement about alcohol.

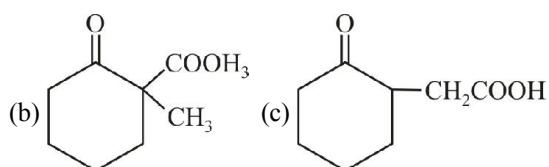
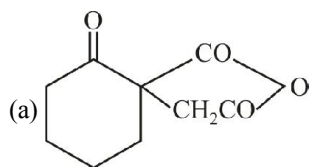
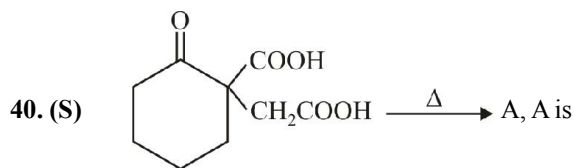
- (a) Configuration about chiral carbon is retained  
(b) Configuration about chiral carbon is inverted  
(c) Product (alcohol) loses chirality  
(d) None of the statements is correct

38. (S)  $CH_3CH_2C(=O)CH_3$  undergoes oxidation by  $CH_3CO_3H$  and then hydrolysed. Products are

- (a)  $CH_3CH_2C(=O)OH$  or  $CH_3OH$   
(b)  $CH_3C(=O)OH$  and  $CH_3CH_2OH$   
(c)  $CH_3CH_2OCH_3$  and  $CH_3OH$   
(d)  $CH_3OCH_3$  and  $CH_3OH$

39. (S)   $\xrightarrow{CH_3OH}$  A, A is

- (a)  $\begin{array}{c} CH_2-CH_2 \\ | \quad | \\ OH \quad COOCH_3 \end{array}$  (b)  $\begin{array}{c} CH_2-CH_2 \\ | \quad | \\ OCH_3 \quad COOH \end{array}$   
(c) Both are correct (d) None is correct

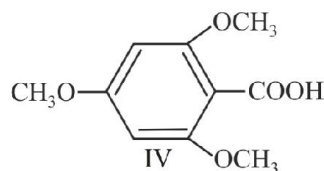
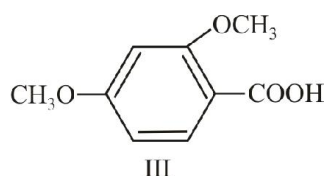
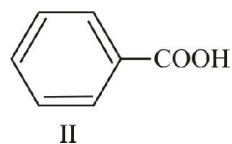
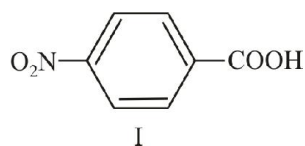


(d) None

41. (S)  $\text{CH}_3\text{Br}$  can be prepared by

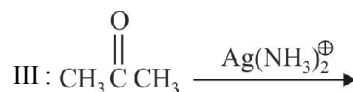
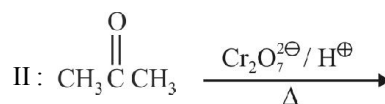
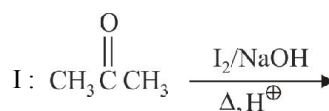
- (a)  $\text{CH}_3\text{COOAg} + \text{Br}_2 \xrightarrow{\Delta}$   
 (b)  $\text{CH}_3\text{COOH} \xrightarrow{\text{P} / \text{Br}_2} \xrightarrow{\text{Soda lime, } \Delta}$   
 (c) Both (d) None

42. (S) Give the order of ease of the esterification of the following acids



- (a)  $\text{I} > \text{II} > \text{III} > \text{IV}$  (b)  $\text{IV} > \text{III} > \text{II} > \text{I}$   
 (c)  $\text{II} > \text{I} > \text{IV} > \text{III}$  (d)  $\text{I} > \text{II} > \text{IV} > \text{III}$

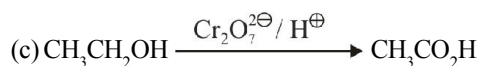
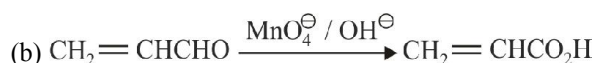
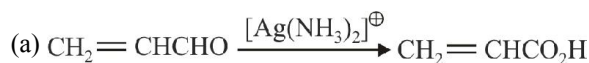
43. (S)  $\text{CH}_3\text{C}(=\text{O})\text{CH}_3$  can be converted to  $\text{CH}_3\text{C}(=\text{O})\text{OH}$  by following methods :



Which methods can be effective ?

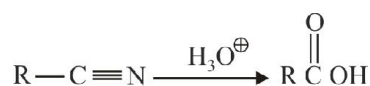
- (a) I, III (b) II, III  
 (c) I, II (d) I, II, III

44. (S) Which reaction does not have the correct oxidising agent ?



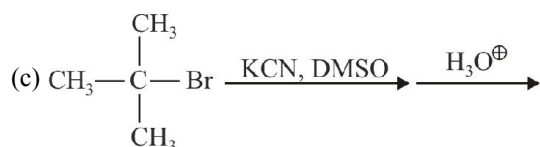
(d) All of the above

45. (S) For the following reaction.



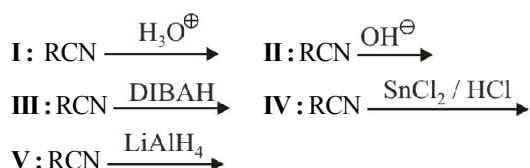
- (a) there is protonation of the electronegative nitrogen followed by attack of water at the electron-deficient carbon  
 (b) an amide is formed as an intermediate  
 (c) nitrogen atom is expelled as ammonium ion  
 (d) all of the above are true

46. (S) In which case yield of the carboxylic acid is maximum ?



(d) Equivalent yield

47. (S) In which cases  $\text{RCO}_2\text{H}$  is not the product ?

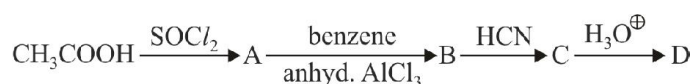


- (a) All except III (b) All except III, V  
 (c) All except I (d) All except I, II

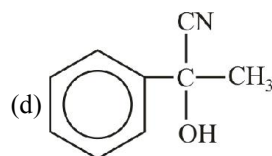
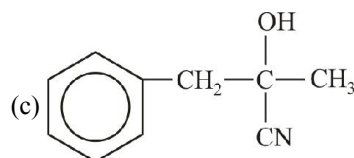
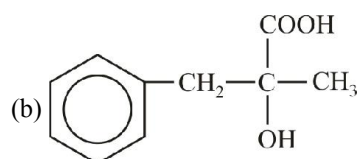
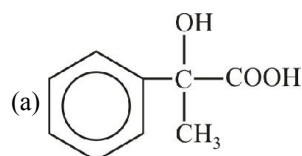
48. (S) Acid hydrolysis of an ester is reversible because :

- (a) alcohol and acid react together to form ester  
 (b) protonation of ester makes carbonyl carbon more reactive for nucleophilic attack  
 (c) proton being strong nucleophile, favours hydrolysis  
 (d)  $\text{H}_3\text{O}^+$  is a nucleophile as well as electrophile

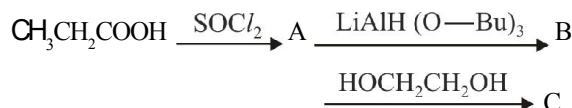
49. (S) In a set of reactions, acetic acid yielded a product D.



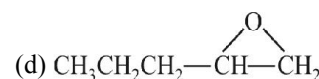
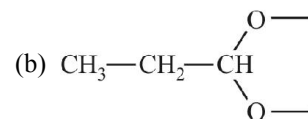
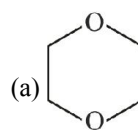
The structure of D would be :



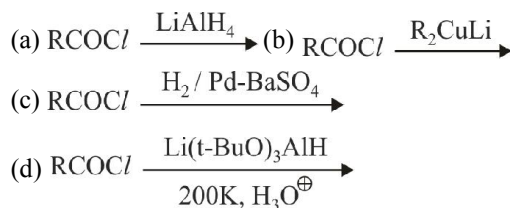
50. (S) In the reaction sequence :



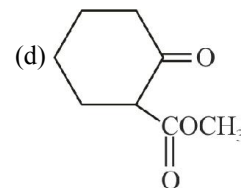
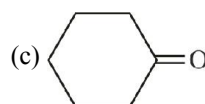
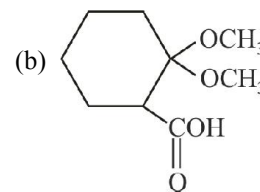
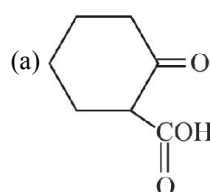
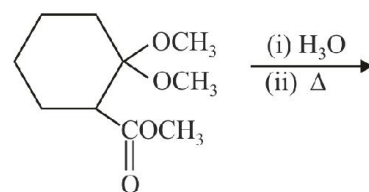
'C' is most likely to be :



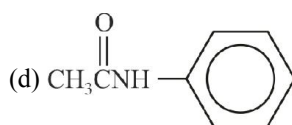
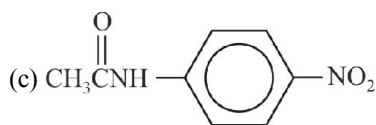
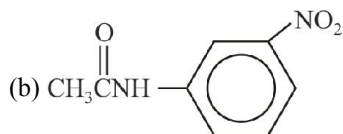
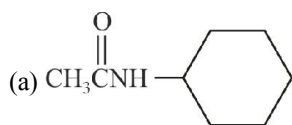
51. (S) Which is not the source of carbonyl compounds ?



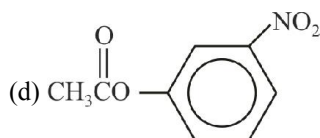
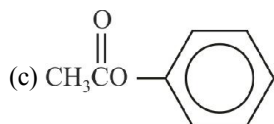
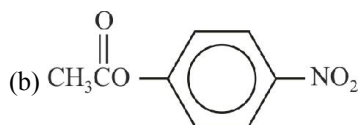
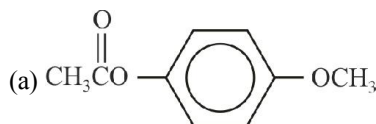
52. (S) What is end product of the following reaction ?



53. (S) Alkaline hydrolysis is maximum in :



54. (S) Acidic hydrolysis is least that of :



55. (A) **Assertion :** Acid catalysed hydrolysis of ester is reversible on the other hand base catalysed hydrolysis is irreversible.

**Reason :** In acid catalysed ester hydrolysis carboxylic acid is formed on which nucleophile attack of alcohol is possible while in base catalysed ester hydrolysis carboxylate anion is formed on which nucleophilic attack is not possible.

- (a) A (b) B  
(c) C (d) D

56. (A) **Assertion :** Acetic acid does not give haloform reaction.

**Reason :** Acetic acid has no  $\alpha$ -hydrogen.

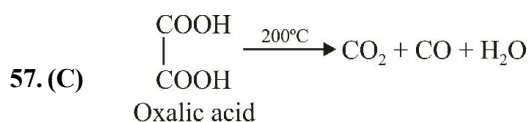
- (a) A (b) B  
(c) C (d) D

### Comprehension

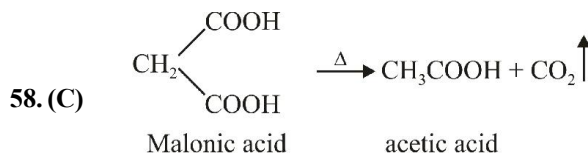
Variety of products are obtained when dicarboxylic acids are heated. The nature of products depends on the carbon chain length separating the two carboxylic groups.

- (i) 1,2 and 1,3-dibasic acids on heating give monocarboxylic acid.  
(ii) 1,4 and 1,5-dibasic acids on heating give cyclic anhydride.  
(iii) 1,6 and 1,7-dibasic acids give cyclic ketones.

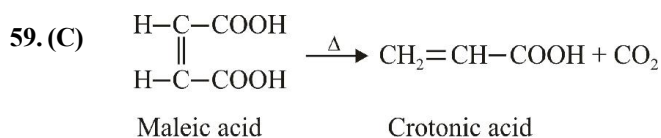
Indicate whether the following reactions are right or wrong :



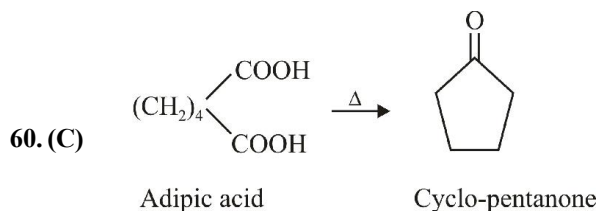
- (a) Right (b) Wrong



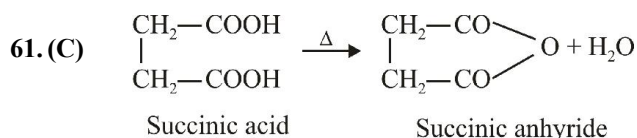
- (a) Right (b) Wrong



- (a) Right (b) Wrong



- (a) Right (b) Wrong

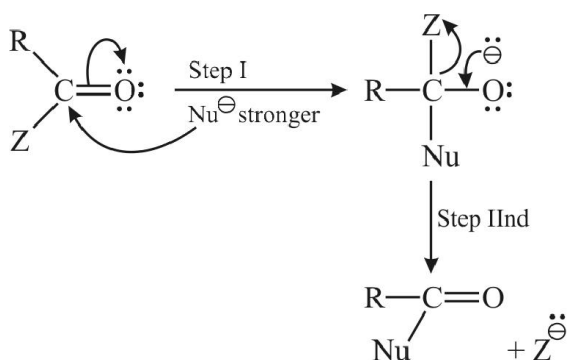


(a) Right

(b) Wrong

### Comprehension

The characteristic and important reaction of carboxylic acid derivatives is nucleophilic acyl substitution which is possible only when the incoming nucleophile ( $\text{Nu}^-$  or  $\text{HNu}$ ) must not be a much weaker base than the group or substituent attached to the acyl group in the acid derivative. It is a two step process involving nucleophilic attack followed by the loss of the leaving group.

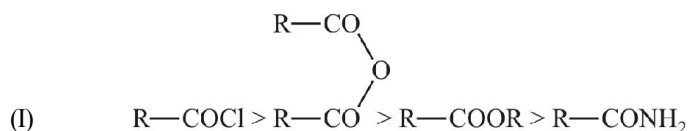


62. (C) Arrange the following in the decreasing order of leaving group ability ?

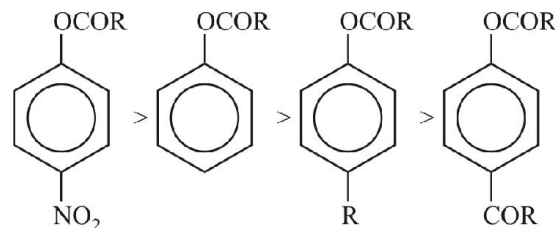
- (I)  $\text{X}^-$  (II)  $\text{NH}_2^-$   
(III)  $\text{OH}^-$  (IV)  $\text{OR}^-$   
(V)  $\text{RCOO}^-$

- (a)  $\text{I} > \text{V} > \text{III} > \text{IV} > \text{II}$  (b)  $\text{I} > \text{V} > \text{III} \approx \text{IV} > \text{II}$   
(c)  $\text{V} > \text{I} > \text{II} > \text{IV} > \text{III}$  (d)  $\text{V} > \text{I} > \text{III} \approx \text{IV} > \text{II}$

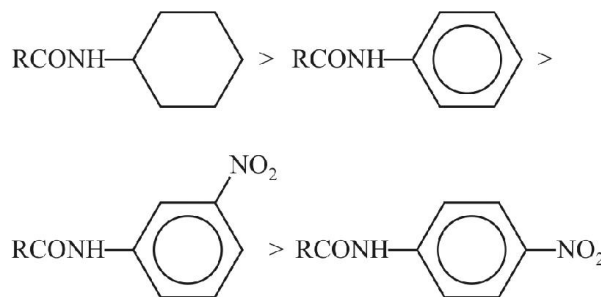
63. (C) Which of these order are correct for hydrolysis ?



(II)



(III)



(a) I, III

(b) II, III

(c) I, II

(d) I, II, III

64. (M) Which of the following reagents will reduce a carboxylic acid to a  $1^\circ$ -alcohol under mild conditions ?

(a)  $\text{BH}_3$ -ether

(b)  $\text{NaBH}_4$ ,  $\text{CH}_3\text{CH}_2\text{OH}$ 

(c)  $\text{H}_2$  and Pt catalyst

(d)  $\text{LiAlH}_4$  in ether

65. (M)  $\text{PhCH}_2\text{CH}_2\text{COOH} \xrightarrow{\text{Ag}_2\text{O}} \text{PhCH}_2\text{CH}_2\text{COOAg} \xrightarrow{\text{Br}_2} \text{PhCH}_2\text{CH}_2\text{Br} + \text{CO}_2 + \text{AgBr}$ . The reaction is known as Hunsdiecker reaction. Choose incorrect statement(s) regarding the Hunsdiecker reaction.

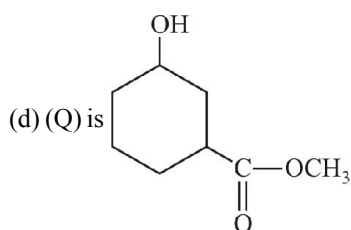
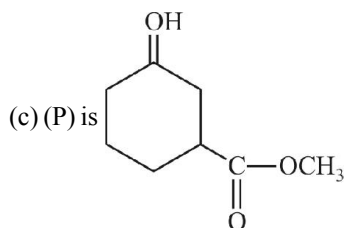
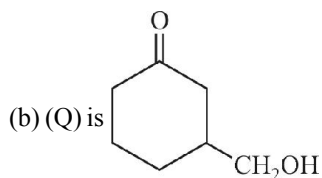
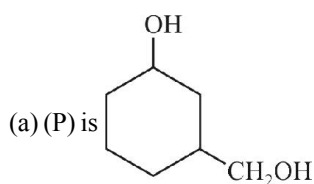
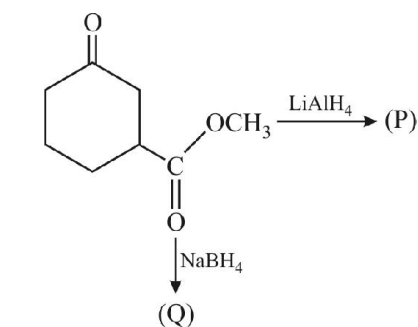
(a) A more convenient way to perform the Hunsdiecker reaction is the use of a mixture of acid and mercuric oxide.

(b) Reaction involves radical intermediates.

(c) The yield of Bromide increases as  $1^\circ$  alkyl  $<$   $2^\circ$  alkyl  $<$   $3^\circ$  alkyl.

(d) The reaction starts with decarboxylation.

66. (M) Identify (P) and (Q) in the following reaction



67. (A) **Assertion :** Formic acid reduces mercuric chloride.

**Reason :** Formic acid reducing aldehydic group.

- (a) A (b) B  
(c) C (d) D

68. (A) **Assertion :** Acetoacetic ester  $\text{CH}_3\text{COCH}_2\text{COOC}_2\text{H}_5$  gives iodoform test.

**Reason :** It has  $\text{CH}_3\text{CO}^-$  group attached to methylene group.

(a) A (b) B

(c) C (d) D

69. (A) **Assertion :**  $(\text{CH}_3)_3\text{C}-\text{COOH}$  does not give HVZ reaction.

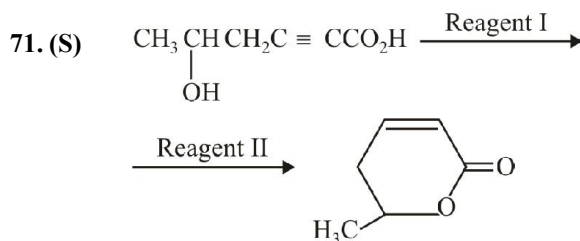
**Reason :** It does not have any  $\alpha$ -hydrogen

- (a) A (b) B  
(c) C (d) D

70. (S) What are the organic products formed



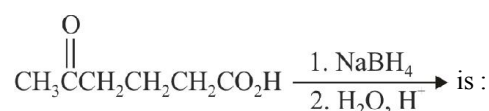
- (a)  $\text{C}_6\text{H}_5-\text{COOH}$  and  $\text{CH}_4$   
(b)  $\text{C}_6\text{H}_5-\text{CH}_2-\text{OH}$  and  $\text{CH}_4$   
(c)  $\text{C}_6\text{H}_5-\text{CH}_3$  and  $\text{CH}_3-\text{OH}$   
(d)  $\text{C}_6\text{H}_5-\text{CH}_2-\text{OH}$  and  $\text{CH}_3-\text{OH}$



Reagents I and II are

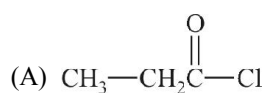
- (a)  $\text{Pt}/\text{H}_2, \text{H}_3\text{O}^+$  (b)  $\text{Li}/\text{NH}_3, \text{H}_3\text{O}^+$   
(c)  $\text{H}_2/\text{Lindlar catalyst}, \text{H}_3\text{O}^+$   
(d)  $\text{H}_2\text{O}/\text{Na}, \text{H}_3\text{O}^+$

72. (S) End product of this conversion :

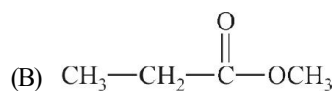


- (a) (b)   
(c) (d)  $\text{CH}_3\underset{\text{OH}}{\text{CH}}\text{CH}_2\text{CH}_2\text{CH}_2\text{CO}_2\text{H}$

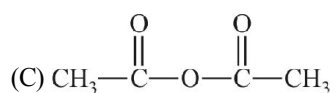
73. (X) Match the following columns



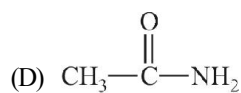
(P) Reaction with  $\text{H}_2\text{O}$   
(at low temperature)



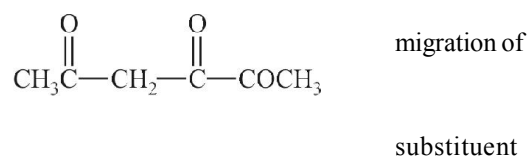
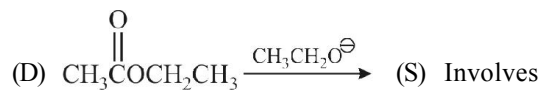
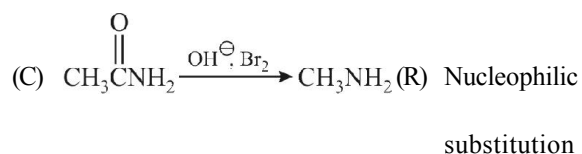
(Q) Alcohol on treatment  
with  $\text{LiAlH}_4$



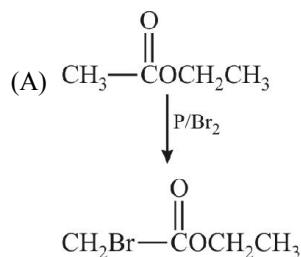
(R) Liberation of  $\text{CH}_4$   
with  $\text{CH}_3\text{MgBr}$



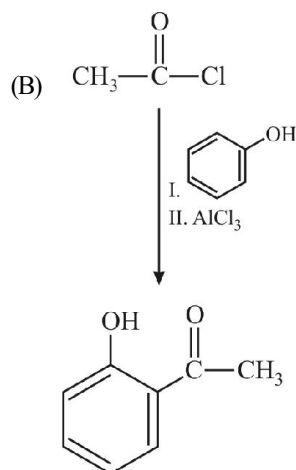
(S) Enolate  
condensation



74. (X) Match the following columns



(P) Intramolecular  
rearrangement



(Q) Electrophilic  
substitution

## EXERCISE - 4 : PREVIOUS YEARS JEE ADVANCED QUESTION

1. Which of the following is basic ? (1980)

- (a)  $\text{CH}_3\text{CH}_2\text{OH}$  (b)  $\text{H}-\text{O}-\text{O}-\text{H}$   
(c)  $\text{HO}-\text{CH}_2-\text{CH}_2-\text{OH}$  (d)  $\text{H}_3\text{C}-\text{C}(=\text{O})-\text{OH}$

2. Acetamide is treated separately with the following reagents. Which one of these would give methyl amine ?

- (a)  $\text{PCl}_5$  (b)  $\text{NaOH} + \text{Br}_2$  (1983)  
(c) Sodalime (d) Hot conc.  $\text{H}_2\text{SO}_4$

3. Hydrogenation of benzoyl chloride in the presence of Pd on  $\text{BaSO}_4$  gives (1992)

- (a) benzyl alcohol (b) benzaldehyde  
(c) benzoic acid (d) phenol

4. When propionic acid is treated with aqueous sodium bicarbonate,  $\text{CO}_2$  is liberated. The C of  $\text{CO}_2$  comes from (1992)

- (a) methyl group (b) carboxylic acid group  
(c) methylene group (d) bicarbonate group

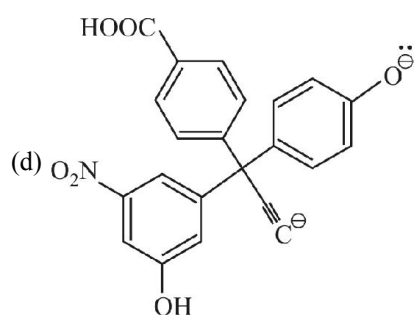
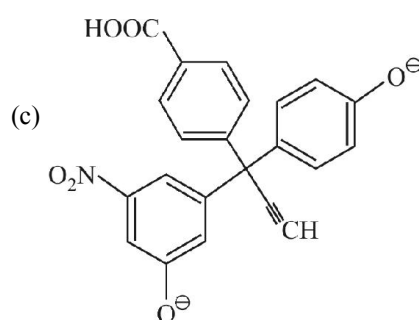
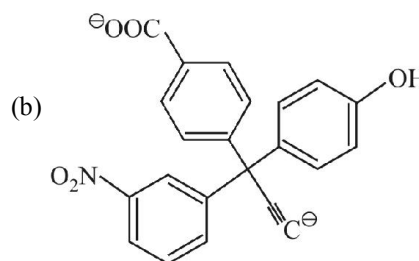
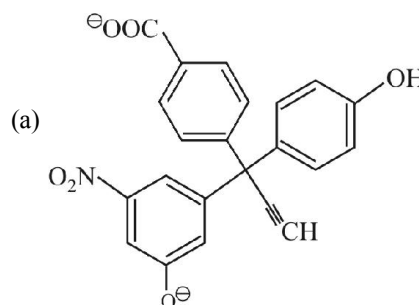
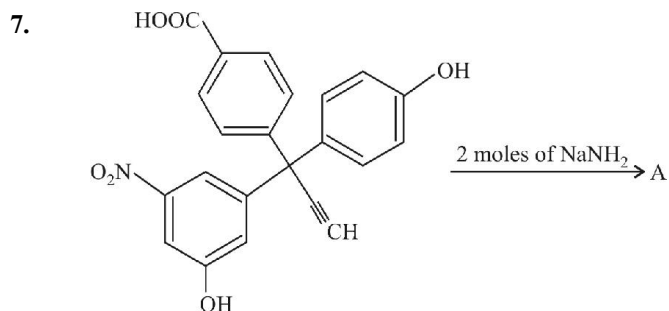
5. Benzoyl chloride is prepared from benzoic acid by

- (a)  $\text{Cl}_2, \text{h}\nu$  (b)  $\text{SO}_2\text{Cl}_2$  (2000)  
(c)  $\text{SOCl}_2$  (d)  $\text{Cl}_2, \text{H}_2\text{O}$

6. Identify the correct order of boiling points of the following compounds (2002)

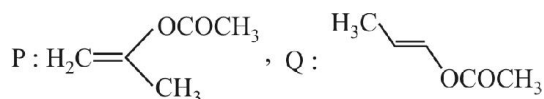
- (A)  $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{OH}$  (B)  $\text{CH}_3\text{CH}_2\text{CH}_2\text{CHO}$   
(C)  $\text{CH}_3\text{CH}_2\text{CH}_2\text{COOH}$

- (a)  $A > B > C$  (b)  $C > A > B$   
(c)  $A > C > B$  (d)  $C > B > A$



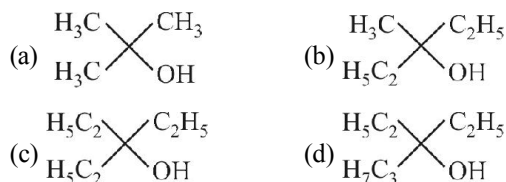
(2003)

8. The product of acid hydrolysis of P and Q can be distinguished by (2003)



- (a) Lucas reagent                      (b) 2, 4-DNP  
(c) Fehling's solution                (d)  $\text{NaHSO}_3$

9. Ethyl ester  $\xrightarrow[\text{(excess)}]{\text{CH}_3\text{MgBr}}$  P, the product 'P' will be (2003)



10. An enantiomerically pure acid is treated with racemic mixture of an alcohol having one chiral carbon. The ester formed will be **(2003)**

- (a) optically active mixture      (b) pure enantiomer  
(c) meso compound                  (d) racemic mixture

**11.** Benzamide on treatment with  $\text{POCl}_3$  gives **(2004)**

- (a) aniline                      (b) benzonitrile  
(c) chlorobenzene            (d) benzyl amine

12. When benzene sulphonic acid and p-nitrophenol are treated with  $\text{NaHCO}_3$ , the gases released respectively, are

- (a)  $\text{SO}_2, \text{NO}_2$  (b)  $\text{SO}_2, \text{NO}$   
(c)  $\text{SO}_2, \text{CO}_2$  (d)  $\text{CO}_2, \text{CO}_2$  **(2006)**

13. In the reaction,  $R-X \xrightarrow{\text{Alcoholic KCN}} A \xrightarrow{\text{Dilute HCl}} B$   
The product B is (2008)

- (a) alkyl chloride                      (b) aldehyde  
(c) carboxylic acid                      (d) ketone

### Assertion Reason

**14. Assertion :** Acetic acid does not undergo haloform reaction. (1998)

**Reason :** Acetic acid has no alpha hydrogen.

- (a) A                  (b) B  
(c) C                 (d) D

15. **Assertion :** p-hydroxybenzoic acid has a lower boiling point than o-hydroxybenzoic acid.

**Reason** : o-hydroxybenzoic acid has intramolecular hydrogen bonding. **(2007)**

- (a) A
- (c) C
- (b) B
- (d) D

16. Reaction of  $\text{R}-\overset{\text{O}}{\parallel}{\text{C}}-\text{NH}_2$  with a mixture of  $\text{Br}_2$  and  $\text{KOH}$  gives  $\text{R}-\text{NH}_2$  as the main product. The intermediates involved in this reaction are **(1992)**

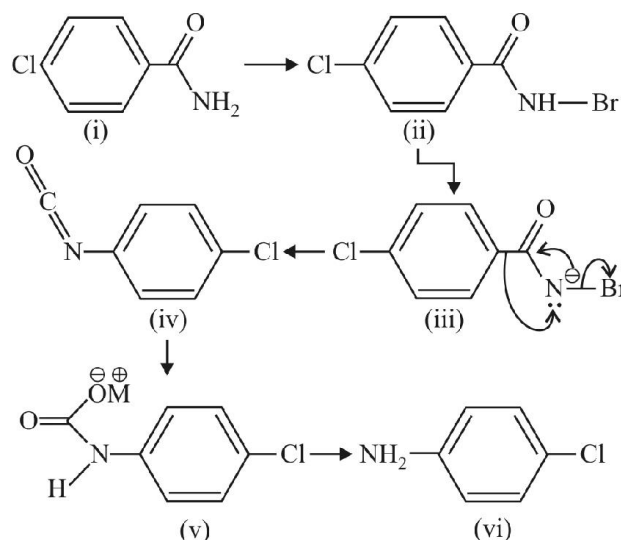
- (a)  $\text{R}-\overset{\text{O}}{\parallel}{\text{C}}-\text{NHBr}$       (b)  $\text{R}-\text{NHBr}$
- (c)  $\text{R}-\text{N}=\text{C}=\text{O}$       (d)  $\text{R}-\overset{\text{O}}{\parallel}{\text{C}}-\text{N} \begin{cases} \text{Br} \\ \text{Br} \end{cases}$

## Comprehension

**Using the following Passage, solve Q. 17 to Q. 19**

### Passage – 1

RCONH<sub>2</sub> is converted into RNH<sub>2</sub> by means of Hofmann bromamide degradation. **(2006)**



In this reaction, RCONHBr is formed from which this reaction has derived its name. Electron donating group at phenyl activates the reaction. Hofmann degradation reaction is an intramolecular reaction.

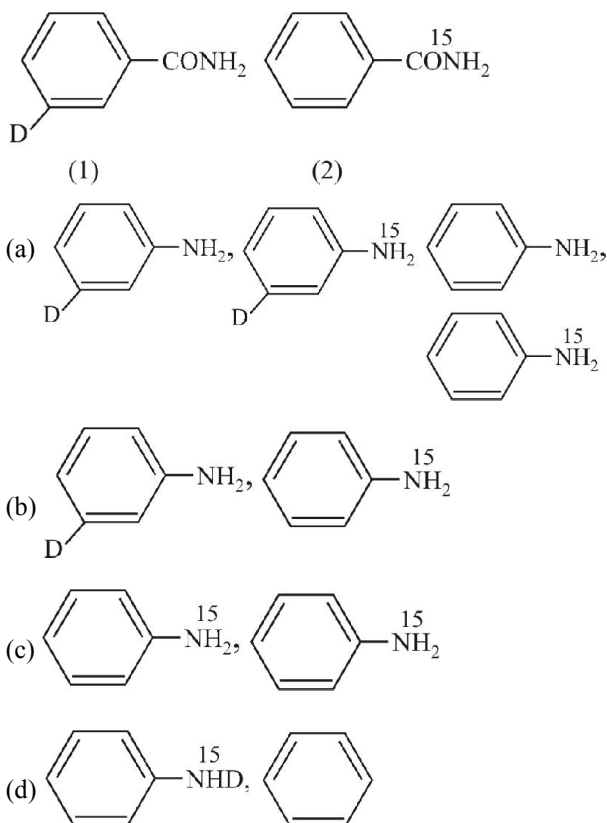
**17.** How can the conversion of (i) to (ii) be brought about?

- (a)  $\text{KBr}$  (b)  $\text{KBr} + \text{CH}_3\text{ONa}$   
(c)  $\text{KBr} + \text{KOH}$  (d)  $\text{Br}_2 + \text{KOH}$

18. Which is the rate determining step in Hofmann bromamide degradation ?

- (a) Formation of (i) (b) Formation of (ii)  
(c) Formation of (iii) (d) Formation of (iv)

19. What are the constituent amines formed when the mixture of (1) and (2) undergoes Hofmann bromamide degradation ?

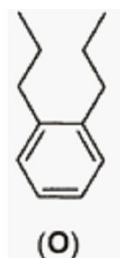


Using the following Passage, solve Q. 20 to Q. 21

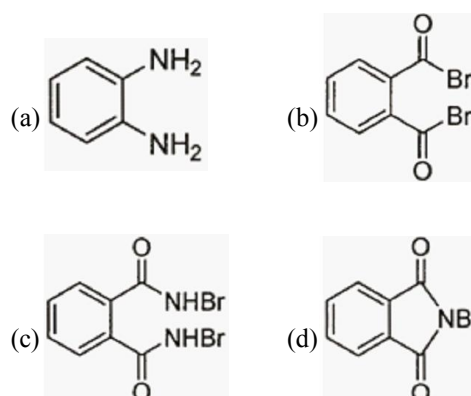
#### PARAGRAPH-2

Treatment of compound **O** with  $\text{KMnO}_4/\text{H}^+$  gave **P**, which on heating with ammonia gave **Q**. The compound **Q** on treatment with  $\text{Br}_2/\text{NaOH}$  produced **R**. On strong heating, **Q** gave **S**, which on further treatment with ethyl 2-bromopropanoate in the presence of  $\text{KOH}$  followed by acidification, gave a compound **T**.

(2016)



20. The compound **R** is



21. The compound **T** is

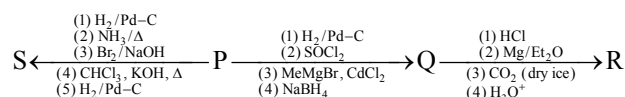
- (a) glycine (b) alanine  
(c) valine (d) serine

Using the following Passage, solve Q. 22 to Q. 23

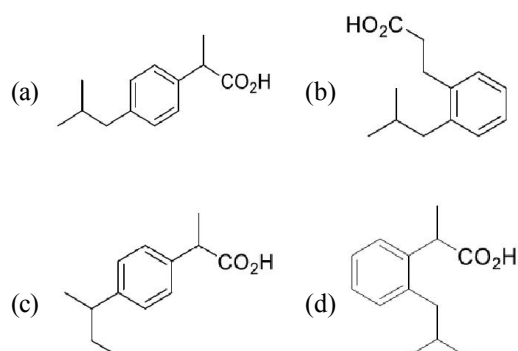
#### PARAGRAPH-3

An organic acid **P** ( $\text{C}_{11}\text{H}_{12}\text{O}_2$ ) can easily be oxidized to a dibasic acid which reacts with ethyleneglycol to produce a polymer dacron. Upon ozonolysis, **P** gives an aliphatic ketone as one of the products. **P** undergoes the following reaction sequences to furnish **R** via **Q**. The compound **P** also undergoes another set of reactions to produce **S**.

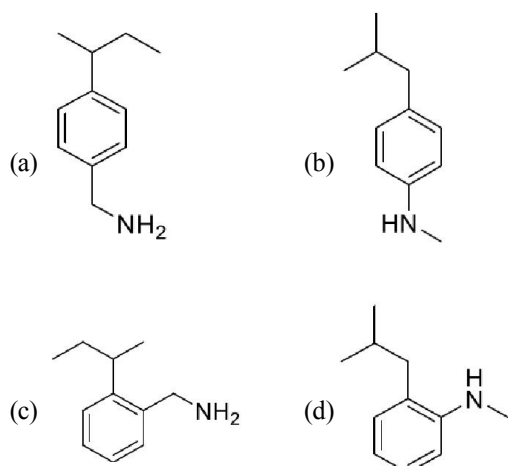
(2018)



22. The compound **R** is



23. The compound S is



### Fill in the Blanks

24. Formic acid when heated with conc  $\text{H}_2\text{SO}_4$  produces ..... (1991)

### True / False

25. Hydrolysis of an ester in presence of a dilute acid is known as saponification. (1983)

26. The boiling point of propionic acid is less than that of n-butyl alcohol, an alcohol of comparable molecular weight. (1991)

### Analytical & Descriptive

27. Write the chemical equation to show what happens when, "Ethyl acetate is treated with sodium ethoxide in ethanol and the reaction mixture is acidified". (1981)

28. Write the structural formula of main organic product formed when ethyl acetate is treated with double the molar quantity of methyl magnesium bromide and the reaction mixture is poured into water. (1982)

29. State with balanced equation, what happens when, "Acetic anhydride reacts with phenol in presence of a base. (1982)

30. Give reasons for the following in one or two sentences. "Acetic acid can be halogenated in the presence of P and  $\text{Cl}_2$ , but formic acid cannot be halogenated in the same way". (1983)

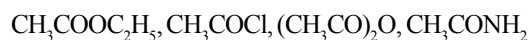
31. Write down the reaction involved in the preparation of the following using the reagents indicated against in parenthesis. (1984)

"Propionic anhydride from propionaldehyde"

$[\text{AgNO}_3, \text{NH}_4\text{OH}, \text{P}_2\text{O}_5]$

32. Give reason in one or two sentences for the following :  
"Formic acid is a stronger acid than acetic acid". (1985)

33. Arrange the following in order of their increasing ease of hydrolysis : (1986)



34. Complete the following with appropriate structures :



35. A liquid X, having a molecular formula  $\text{C}_6\text{H}_{12}\text{O}_2$  is hydrolysed with water in the presence of an acid to give a carboxylic acid Y and an alcohol Z. Oxidation of Z with chromic acid give Y. What are the structures of X, Y and Z. (1986)

36. Write balanced equations for the following reaction :

"Acetamide is reacted with bromine in the presence of potassium hydroxide". (1987)

37. Give reasons for : (1988)

"Carbon-oxygen bond lengths in formic acid are 1.23 Å and 1.36 Å and both the carbon-oxygen bonds in sodium formate have the same value, ie, 1.27 Å".

38. Arrange the following as stated : (1991)

"Increasing order of acidic strength".

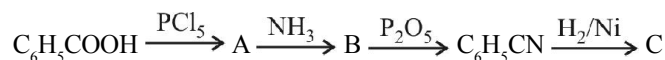
(a) (b) (c)



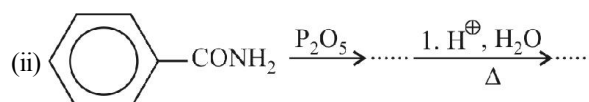
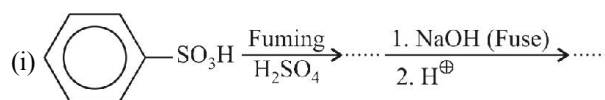
(d) (e)



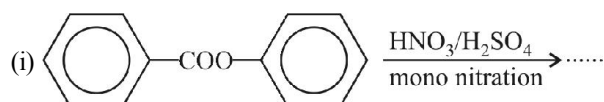
39. In the following identify the compounds/reaction conditions represented by the alphabets A, B, and C : (1991)



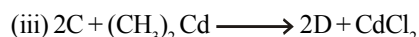
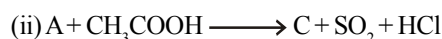
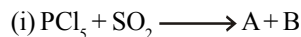
40. Complete the following sequence of the reactions with appropriate structures (1992)



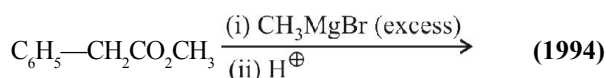
41. Identify the major product in the following reactions :



42. In the following reactions identify the compounds A, B, C and D. (1994)



43. Predict the major product in the following reaction :



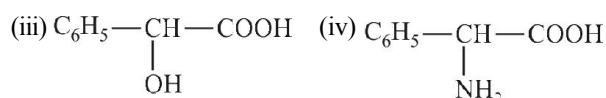
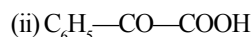
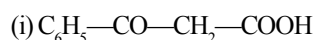
44. How will you bring about the following conversion ?

“Benzamide from nitrobenzene”. (1994)

45. Give reason for the following

“In acylium ion the structure  $\text{R—C}\equiv\text{O}^+$  is more stable than  $\text{R—C}^+=\text{O}$ ” (1994)

46. Which of the following carboxylic acids undergoes decarboxylation easily ? Explain briefly. (1995)

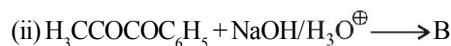
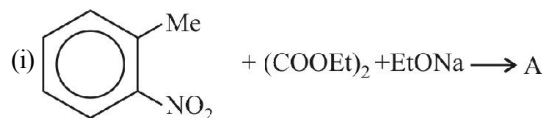


47. Complete the following sequence of reactions with appropriate structures (1995)



48. A hydrocarbon A of the formula  $\text{C}_8\text{H}_{10}$ , on ozonolysis gives compound B ( $\text{C}_4\text{H}_6\text{O}_2$ ) only. The compound B can also be obtained from the alkyl bromide C ( $\text{C}_3\text{H}_5\text{Br}$ ) upon treatment with magnesium in dry ether, followed by carbon dioxide and acidification. Identify A, B and C and also give equations for the reactions. (1996)

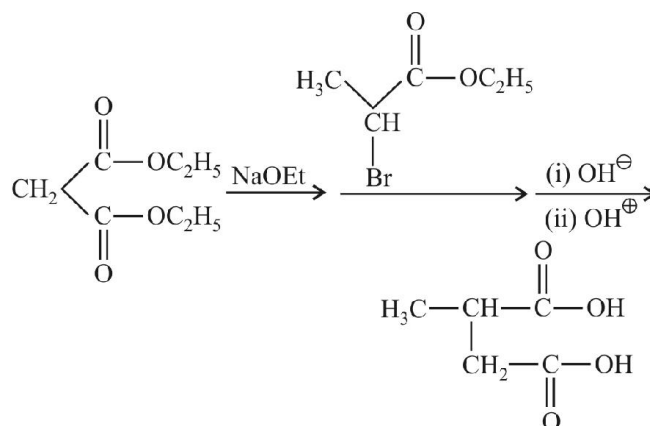
49. Complete the following giving the structures of the principal organic products (1997)



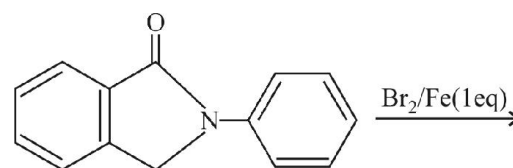
50. Following reaction gives two products. Write the structures of the products. (1998)



51. Explain briefly the formation of the products giving the structures of the intermediates (1999)



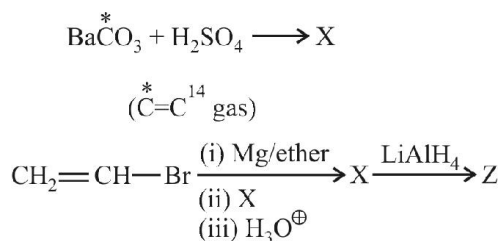
52. What would be the major product in the following reactions ? (2000)



53. Write the structures of the products A and B. (2000)



54. Identify X and Y in the following synthetic scheme and write their structures. Explain the formation of labelled formaldehyde ( $\text{H}_2\text{CO}$ ) as one of the products when compound (Z) is treated with HBr and subsequently ozonolysed. Mark the  $\text{C}^*$  carbon in the entire scheme. (2001)

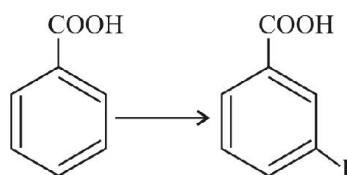


55. Five isomeric para-disubstituted aromatic compounds A to E with molecular formula  $\text{C}_8\text{H}_8\text{O}_2$  were given for identification. (2002)

Based on the following observations, given structures of the compounds :

- Both A and B form a silver mirror with Tollen's reagent; also, B gives a positive test with  $\text{FeCl}_3$  solution.
- C gives positive iodoform test.
- D is readily extracted in aqueous  $\text{NaHCO}_3$  solution.
- E on acid hydrolysis gives, 1, 4-dihydroxy benzene.

56. A racemic mixture of  $(\pm)$  2-phenyl propanoic acid on esterification with  $(\pm)$  2-butanol gives two esters. Mention the stereochemistry of the two esters produced. (2003)
57. Convert (in not more than three steps) (2003)



58. Compound A of molecular formula  $\text{C}_9\text{H}_7\text{O}_2\text{Cl}$  exists in keto form and predominantly in enolic form B. On oxidation with  $\text{KMnO}_4$ , A gives m-chlorobenzoic acid. Identify A and B. (2003)

## Answer Key

### EXERCISE - 1 (BASIC OBJECTIVE QUESTIONS)

- |         |         |         |         |         |         |         |         |         |         |
|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|
| 1. (d)  | 2. (c)  | 3. (b)  | 4. (d)  | 5. (c)  | 6. (c)  | 7. (c)  | 8. (c)  | 9. (d)  | 10. (d) |
| 11. (b) | 12. (b) | 13. (c) | 14. (c) | 15. (c) | 16. (c) | 17. (a) | 18. (d) | 19. (c) | 20. (c) |
| 21. (d) | 22. (d) | 23. (c) | 24. (a) | 25. (d) | 26. (a) | 27. (c) | 28. (c) | 29. (d) | 30. (a) |
| 31. (a) | 32. (d) | 33. (b) | 34. (c) | 35. (d) | 36. (a) | 37. (d) | 38. (d) | 39. (a) | 40. (a) |
| 41. (b) | 42. (b) | 43. (a) | 44. (b) | 45. (d) | 46. (b) | 47. (c) | 48. (b) | 49. (c) | 50. (b) |
| 51. (a) | 52. (c) | 53. (c) | 54. (d) | 55. (c) | 56. (d) | 57. (c) | 58. (a) | 59. (a) | 60. (d) |
| 61. (c) | 62. (c) | 63. (b) |         |         |         |         |         |         |         |

### EXERCISE - 2 (PREVIOUS YEAR JEE MAINS QUESTIONS)

- |         |         |         |         |         |        |        |        |        |         |
|---------|---------|---------|---------|---------|--------|--------|--------|--------|---------|
| 1. (c)  | 2. (c)  | 3. (a)  | 4. (b)  | 5. (d)  | 6. (c) | 7. (c) | 8. (c) | 9. (d) | 10. (c) |
| 11. (b) | 12. (b) | 13. (b) | 14. (c) | 15. (d) |        |        |        |        |         |

#### JEE Mains Online

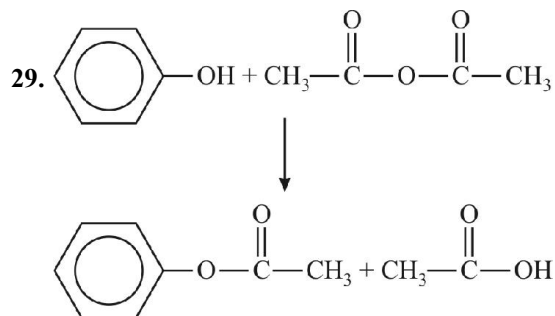
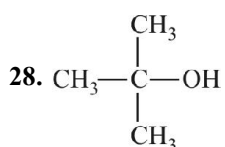
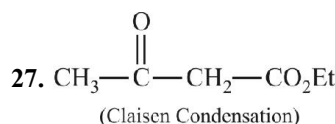
- |        |        |        |        |        |        |        |
|--------|--------|--------|--------|--------|--------|--------|
| 1. (a) | 2. (b) | 3. (c) | 4. (b) | 5. (d) | 6. (c) | 7. (c) |
|--------|--------|--------|--------|--------|--------|--------|

### EXERCISE - 3 (ADVANCED OBJECTIVE QUESTIONS)

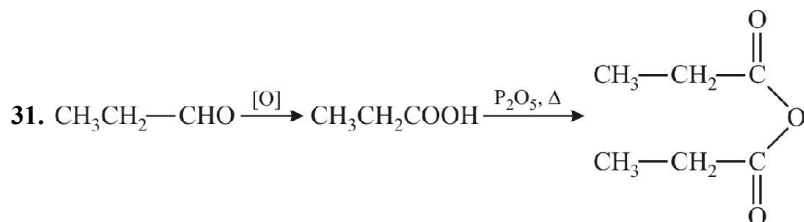
- |             |         |                                                  |           |           |           |                                               |         |         |             |
|-------------|---------|--------------------------------------------------|-----------|-----------|-----------|-----------------------------------------------|---------|---------|-------------|
| 1. (d)      | 2. (b)  | 3. (a)                                           | 4. (b)    | 5. (b)    | 6. (b)    | 7. (a)                                        | 8. (d)  | 9. (d)  | 10. (d)     |
| 11. (a)     | 12. (d) | 13. (d)                                          | 14. (b)   | 15. (a)   | 16. (d)   | 17. (a)                                       | 18. (d) | 19. (b) | 20. (a,b,c) |
| 21. (a,c,d) | 22. (a) | 23. (d)                                          | 24. (c)   | 25. (d)   | 26. (b)   | 27. (c)                                       | 28. (c) | 29. (c) | 30. (d)     |
| 31. (c)     | 32. (d) | 33. (c)                                          | 34. (c)   | 35. (a)   | 36. (a)   | 37. (a)                                       | 38. (b) | 39. (a) | 40. (c)     |
| 41. (c)     | 42. (a) | 43. (c)                                          | 44. (b)   | 45. (d)   | 46. (a)   | 47. (d)                                       | 48. (d) | 49. (a) | 50. (b)     |
| 51. (a)     | 52. (c) | 53. (c)                                          | 54. (a)   | 55. (a)   | 56. (c)   | 57. (a)                                       | 58. (a) | 59. (b) | 60. (a)     |
| 61. (a)     | 62. (b) | 63. (c)                                          | 64. (a,d) | 65. (c,d) | 66. (a,d) | 67. (a)                                       | 68. (d) | 69. (a) | 70. (d)     |
| 71. (c)     | 72. (a) | 73. A – P, Q; B – P, Q, S; C – P, Q, S; D – P, R |           |           |           | 74. A – Q; B – P, Q, R, S; C – P, Q, S; D – R |         |         |             |

## EXERCISE - 4 (PREVIOUS YEAR JEE ADVANCED QUESTIONS)

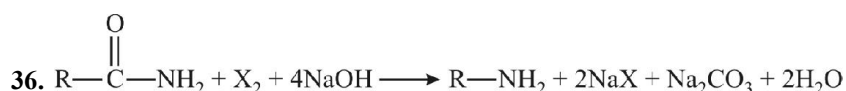
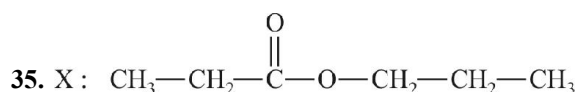
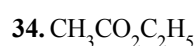
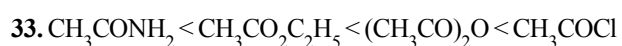
1. (a)      2. (b)      3. (b)      4. (d)      5. (c)      6. (b)      7. (a)      8. (c)      9. (a)
10. (a)      11. (b)      12. (d)      13. (c)      14. (c)      15. (d)      16. (a, c)      17. (d)      18. (d)
19. (b)      20. (a)      21. (b)      22. (a)      23. (b)
24.  $\text{CO} + \text{H}_2\text{SO}_4 \cdot \text{H}_2\text{O}$  (conc.  $\text{H}_2\text{SO}_4$  acts as dehydrating reagent)      25. False      26. False



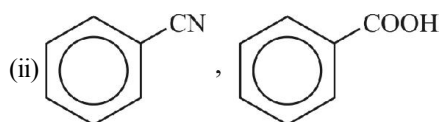
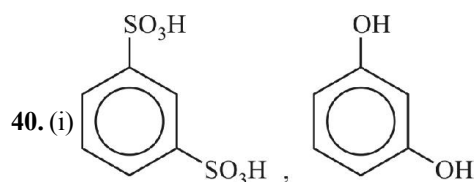
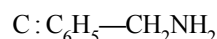
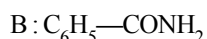
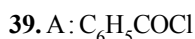
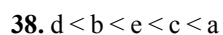
30. Acetic acid has  $\alpha$ -hydrogen but formic acid has no  $\alpha$ -hydrogen.

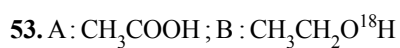
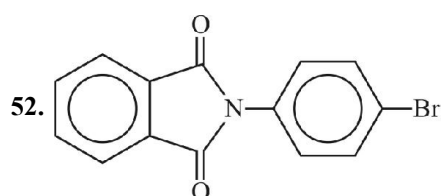
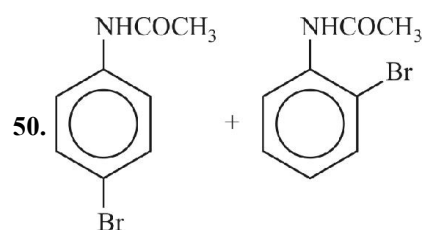
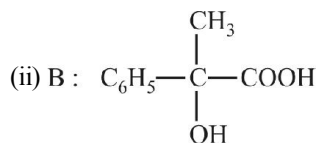
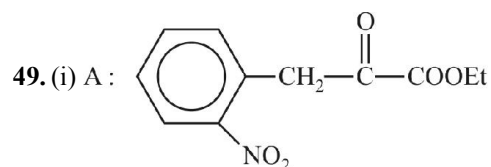
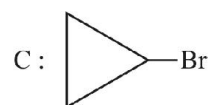
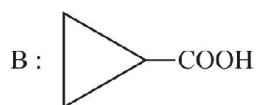
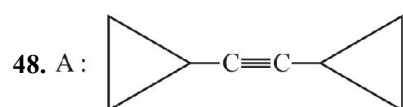
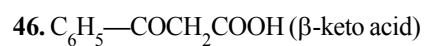
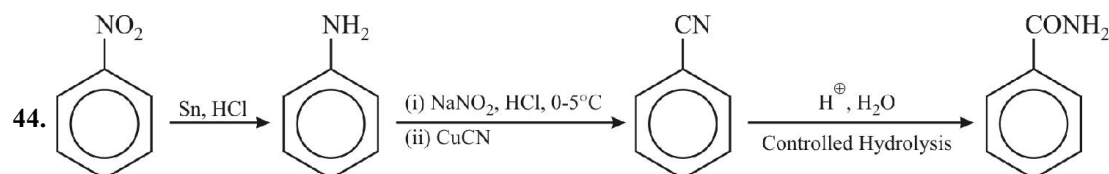
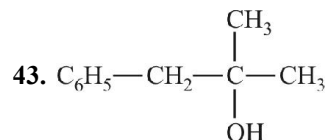
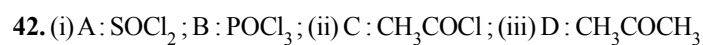
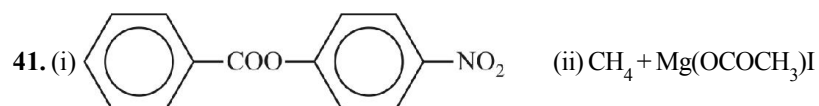


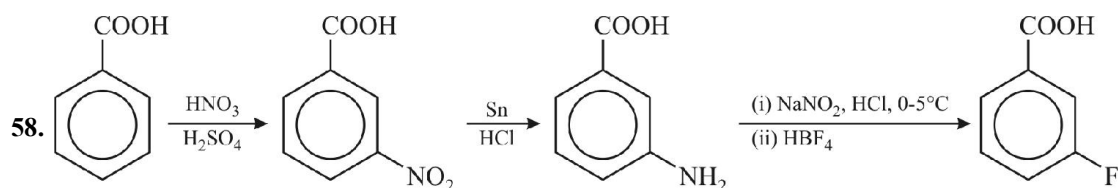
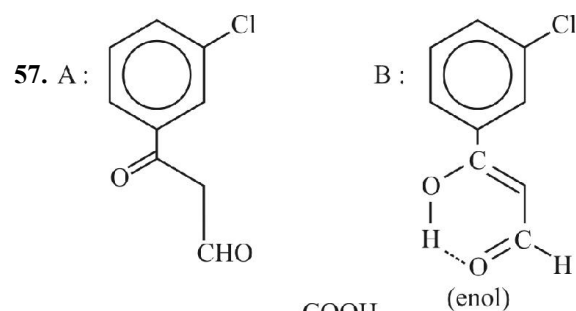
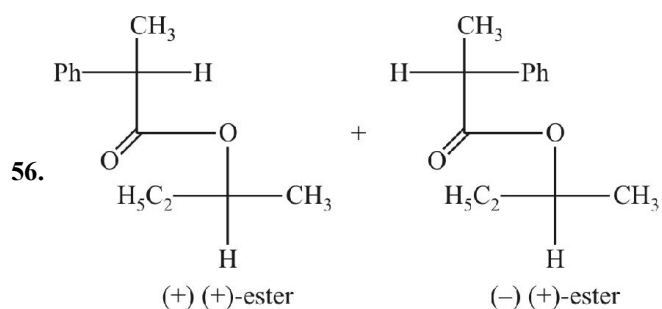
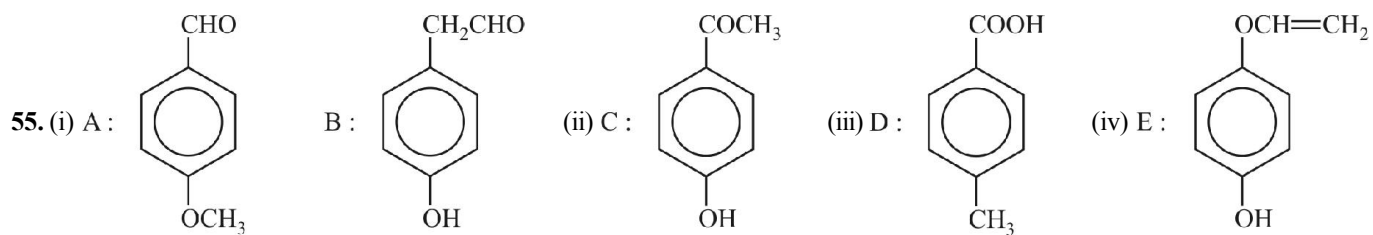
32. Presence of ERG, methyl weakens the acidic strength of  $\text{CH}_3\text{COOH}$ .



37. Equivalent structure in formate ion while one of the resonating structures contributes more in hybrid in formic acid.







*Dream on !!*

